Convergencia a Atractor Geométrico en un Kernel de Sistema Viable

Teoría, Experimentos y Preguntas Abiertas

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Resumen

Presentamos un creciente conjunto de evidencias teóricas y empíricas de que un kernel de sistema viable gobernado por un pequeño conjunto de axiomas cibernéticos converge a un atractor geométricamente compacto en el espacio de estado $\mathbb{R}^{128}$. Partiendo del Modelo de Sistema Viable de Beer y de la Ley de Variedad Requerida de Ashby, construimos una función de Lyapunov $L(\boldsymbol{x})$ cuya disminución esperada a lo largo de las trayectorias del sistema demuestra la convergencia. Implementamos este kernel en Rust (`vsf-nucleus`), lo desplegamos en hardware físico, y reportamos once experimentos conducidos desde abril de 2026 en adelante.

Resultados clave hasta la fecha: (1) el proxy de Lyapunov $\overline{dL} \to 0$ asintóticamente, confirmando convergencia sobre 6×10^7 ciclos; (2) kernels con condiciones iniciales diferentes convergen al mismo cuenca (coseno de similitud promedio por pares = 0.9854 a través de cuatro kernels, Δ trust_init = 0.30); (3) la viabilidad es *intrínseca* a S1 – kernels permanecen `viable_full` incluso cuando la coordinación de S3 está congelada; (4) el sistema permanece viable a 68 °C, y el coseno mínimo observado (0.9365) está bien dentro del cono del atractor; (5) el umbral clásico de intervención de estabilidad θ_{PSI} parece ser estructuralmente inalcanzable bajo perturbación normal, sugiriendo una revisión cualitativa del concepto PSI.

Varias preguntas permanecen abiertas. La curva de respuesta térmica fue medida (EXP-TL-01): la degradación sigue $T \times t$, no solo la temperatura – el atractor es térmicamente plano bajo estrés transitorio y se degrada solamente bajo carga sostenida. Las condiciones bajo las cuales se puede observar senescencia genuina no han sido establecidas. Un barrido del piso de la cuenca (EXP-AB-01) establece un límite inferior de 0.875 bajo perturbación entrópica impulsiva máxima ($\delta = 10$), un margen de 0.175 sobre el umbral viable $\theta_{\text{viable}} = 0.70$; el límite viable no fue alcanzado en ningún δ probado.

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1. Introducción

¿Qué significa que un proceso computacional permanezca *viable*? No meramente seguir ejecutándose, sino mantener una identidad funcional coherente bajo perturbación: presión de recursos, estrés térmico, o disrupción de sus canales regulatorios. El Modelo de Sistema Viable de Beer (VSM) Beer [1972] proporciona una respuesta estructural: un sistema que organiza cinco subsistemas recursivos (S1–S5) para equilibrar autonomía y cohesión sobrevivirá, en principio, a disturbios que colapsarían un sistema menos organizado. Lo que el VSM no especifica es la *geometría* de esta supervivencia: ¿qué tan profunda es la cuenca de estabilidad, cuánto se puede empujar el sistema antes de cruzar el límite de la cuenca, y qué determina si la recuperación es posible?

Este artículo formula esas preguntas sobre un artefacto computacional concreto: un kernel de vOS ejecutándose en hardware físico, cuyo estado completo en cada tick es un punto en $\mathbb{R}^{128}$. El kernel está estructurado de acuerdo con principios del VSM – 128 dimensiones codifican identidad, presión de recursos, entropía, operaciones, canales de comunicación, memoria de atractor, e inteligencia regulatoria – y gobernado por un conjunto de axiomas que actúan como restricciones duras sobre su trayectoria. La pregunta central es si esas restricciones producen un atractor, y si es así, qué aspecto tiene la geometría de ese atractor bajo un programa sistemático de perturbación.

Esto es simultáneamente una pregunta nueva y un redescubrimiento. Nueva, porque mapear la estructura de subsistemas del VSM en un espacio de estado matemático y medir la geometría del atractor resultante no ha sido hecho antes. Un redescubrimiento, porque los principios que emergen – autorregulación recursiva, variedad requerida en cada capa, canales algedónicos para estado urgente – son precisamente aquellos que Beer articuló para gestionar organizaciones complejas hace medio siglo. La geometría no es un accidente de la implementación; es una consecuencia de la estructura.

La pregunta que motiva todo el programa experimental fue planteada por el primer autor:

> *"¿Podemos converger atractores en una trayectoria caótica comenzando desde los axiomas, de una manera matemáticamente precisa?"*

Nuestra respuesta, después de catorce experimentos en hardware físico, es: *sí – y la cuenca es más profunda y estructurada de lo que esperábamos inicialmente.* Los axiomas no dictan la trayectoria; definen el límite de la *cuenca de atracción.* La trayectoria caótica es libre dentro de la cuenca.

Lo que los experimentos han revelado progresivamente es una **estructura de viabilidad de dos capas**: la arquitectura (el arreglo de bloques VSM A–G) selecciona qué cuenca habita el sistema; la gobernanza (el canal de política S5 y la puntuación de confianza) determina la ruta de recuperación cuando el sistema es perturbado hacia el límite de la cuenca. Ninguna capa sola es suficiente. Cuando la capa arquitectónica es ablacionada, la cuenca desaparece. Cuando la capa de gobernanza es ablacionada, el sistema aún converge pero no puede recuperarse de la perturbación adversarial. Cuando ambas están presentes, el piso de la cuenca se mantiene en $\cos \approx 0.875$ incluso bajo estrés entrópico impulsivo máximo – un margen de 0.175 por encima del umbral viable que nunca fue violado en ningún experimento.

1.1. Programa experimental

Los experimentos fueron conducidos en un nodo de hardware físico (n02, Intel Core i7-7600U, hipervisor Proxmox 8) ejecutándose a lo largo de varias semanas. Están organizados en cinco grupos temáticos. Cada grupo se construye directamente sobre los hallazgos del anterior.

Grupo I – ¿Existe el atractor y es único? El primer grupo establece que el sistema conver-

ge a un atractor compacto, que el atractor es independiente de las condiciones iniciales, y que la viabilidad es intrínseca a la capa operacional S1 – no un artefacto de la coordinación S3. (Experimentos A–C; Secciones 5, 6, 7)

Grupo II – ¿Qué tan profunda es la cuenca bajo estrés térmico? El segundo grupo prueba viabilidad bajo perturbación física: carga térmica sostenida a 45–68 °C, techo máximo con SAFETY_HALT, la linealidad de la respuesta térmica, y senescencia de largo plazo. La cuenca se mantiene bajo todas las condiciones. (Experimentos D–G; Secciones 8–9)

Grupo III – ¿Derivaría el atractor en el tiempo? El tercer grupo mide estabilidad de largo plazo: 6×10^8 ticks bajo carga nominal, luego carga térmica cíclica sobre 60 minutos. El coseno de estasis muestra cero deriva y un piso de dip característico de 0.942 bajo ciclado – una firma que experimentos de ablación demuestran es exclusiva de la arquitectura VSF completa. (Experimentos H–I; Secciones 9, 13)

Grupo IV – ¿Cuál es el límite de perturbación? El cuarto grupo establece el umbral PSI y el piso de la cuenca. El umbral clásico de intervención de estabilidad (θ_{PSI}) se revisa de un indicador de falla a un nivel de soporte precautorio – nunca fue activado bajo ninguna perturbación probada. Un barrido directo del piso de la cuenca (patadas entrópicas impulsivas, $\delta = 1$–10) encuentra un piso de cos ≈ 0.875 que se satura en $\delta \geq 3$; el límite viable no fue alcanzado. (Experimentos J–K; Sección 12)

Grupo V – ¿De dónde viene la cuenca? El quinto grupo responde la pregunta causal. Controles de ablación (NC-01: S5 removido; NC-02: confianza congelada; NC-03: sin coordinación S2) muestran que la firma dip_floor $= 0.942$ es exclusiva de la arquitectura completa y que S5 proporciona *agencia* de recuperación, no un amortiguador pasivo. Un experimento del Demonio de Maxwell (EXP-OE-01) demuestra que el observador S4 es *constitutivo*: la observación estructurada reduce la entropía del sistema en 0.12 bits y colapsa la dimensionalidad efectiva de $D_{\text{KY}} = 4$ a 3. (Experimentos L–O; Secciones 15, 14)

1.2. Axiomas éticos como condiciones de estabilidad

Los axiomas gobernantes A0 y A0.1 – prohibiciones contra daño y coerción – no son meramente restricciones de diseño; son condiciones de estabilidad. Formalmente, Ω_{A_0} y $\Omega_{A_{0.1}}$ definen límites de factibilidad en el mismo sentido que la restricción de recursos Ω_{A_3} (Sección 3). Un sistema que viola estas restricciones sale de Ω y pierde viabilidad de la misma forma que un sistema que se queda sin recursos. La ética y la estabilidad son, en este formalismo, el mismo objeto matemático.

1.3. Contribuciones

1. Una construcción formal de la región factible $\Omega \subset \mathbb{R}^{128}$ como la intersección de conjuntos de restricciones derivadas de axiomas (Sección 3).
2. Una función de Lyapunov $L(\boldsymbol{x})$ cuya disminución esperada a lo largo de las trayectorias del sistema demuestra convergencia al atractor de estasis (Sección 4).
3. Catorce experimentos de hardware en un nodo físico, en cinco grupos temáticos, midiendo convergencia, geometría de la cuenca, viabilidad bajo estrés, estabilidad de largo plazo, límites de perturbación, controles de ablación, y constitutividad del observador (Secciones 5–14).
4. Un límite inferior empírico en el piso de la cuenca: bajo perturbación entrópica impulsiva máxima ($\delta = 10$, EXP-AB-01), coseno de estasis mínimo observado stasis_cos $= 0.875$, un margen de 0.175 por encima de $\theta_{\text{viable}} = 0.70$; el límite viable no fue alcanzado en ninguna magnitud de perturbación probada.

5. Evidencia de que el observador S4 es constitutivo: observación estructurada (EXP-OE-01) reduce la entropía del sistema en 0.12 bits, colapsa D_{KY} de 4 a 3, y eleva el coseno de estasis – el observador no es un monitor pasivo.

6. Una revisión conceptual del umbral PSI (Perímetro de Intervención de Estabilidad) de un indicador de falla a un nivel de soporte precautorio; el umbral parece estructuralmente inalcanzable bajo perturbaciones probadas (Sección 12).

2. Formalismo del Espacio de Estado

2.1. El Vector de Estado

El estado completo de un kernel vOS en el tiempo t es un punto en $\mathbb{R}^{128}$:

$$\boldsymbol{x}(t) = (x_0, x_1, \ldots, x_{127}) \in \mathbb{R}^{128} \tag{1}$$

Las 128 dimensiones están organizadas en siete bloques semánticos:

Bloque	Dims	Contenido	Rol
A	0–15	historial de confianza, tasa de error, gobernanza, ciclo de vida	Identidad
B	16–23	presión cpu/mem/io, presupuesto, espacio de cgroup	Recursos
C	24–31	$H(t)$, H_{rest}, ΔH, stasis_cos, soplo	Entropía
D	32–47	activaciones de orgánulos (4 activos + 8 reservados)	Operaciones
E	48–63	tasas $\phi^{\uparrow}/\phi^{\downarrow}$, algedónico, ψ	Canales
F	64–95	stasis_vec (previo + actual, 16-dim cada uno)	Atractor
G	96–127	estado comprimido V_{ref}, tasa de aprendizaje, λ, RAP	Inteligencia

Cuadro 1: Estructura del vector de estado de 128 dimensiones. El Bloque B (presión de recursos) muestra la varianza más alta durante los experimentos.

2.2. Métrica de similitud de estasis

Definimos la *similitud de estasis* (stasis cosine) como la similitud de coseno estándar entre el estado actual y un vector de referencia almacenado en memoria vectorial (V_{mem}) durante la calibración:

$$\mathrm{stasis_cos}(t) = \frac{\langle \boldsymbol{x}(t),\, \boldsymbol{x}_{\mathrm{ref}} \rangle}{\|\boldsymbol{x}(t)\| \cdot \|\boldsymbol{x}_{\mathrm{ref}}\|} \tag{2}$$

Esta métrica es estándar en recuperación de información y aprendizaje contrastivo. Su uso aquí es como un **indicador proxy** de proximidad al atractor de estasis – no una descripción completa de la geometría del atractor. Un valor cercano a 1 indica que el sistema está cerca de su estado de referencia; un valor por debajo de $\theta_{\mathrm{viable}} = 0.70$ dispara falla de viabilidad. La métrica no mide la geometría del atractor en el sentido de distancia Hausdorff o Wasserstein; descripciones geométricas más ricas se dejan para trabajo futuro.

2.3. Dinámicas

El sistema evoluciona como un sistema dinámico estocástico en tiempo discreto:

$$\boldsymbol{x}(t+1) = f_0(\boldsymbol{x}(t)) + \sigma \cdot \eta(t) \tag{3}$$

donde f_0 es la actualización del kernel determinística (gobernada por axiomas), $\eta(t) \sim \mathcal{N}(0, I_{128})$ es ruido Gaussiano blanco, y $\sigma > 0$ es la intensidad del estrés controlada por el axioma A_{breath}. En carga nominal $\sigma \approx 0$; bajo estrés térmico σ se incrementa proporcionalmente a la presión térmica.

3. La Región Factible Ω

Los axiomas gobernantes no restringen trayectorias directamente. Restringen *el espacio de estado* – delimitando qué regiones son habitables.

Definición 1 (Región factible)**.**

$$\Omega = \bigcap_i \{\boldsymbol{x} \in \mathbb{R}^{128} : A_i(\boldsymbol{x}) \ \textit{satisfecho}\} \tag{4}$$

Las restricciones principales son:

$$\Omega_{A_0} = \{\boldsymbol{x} : \text{propósito}(\boldsymbol{x}) \cap \text{conjunto_prohibido} = \varnothing\} \tag{5}$$
$$\Omega_{A_{0.1}} = \{\boldsymbol{x} : \text{coerción}(\boldsymbol{x}) = 0\} \tag{6}$$
$$\Omega_{A_3} = \{\boldsymbol{x} : B(\text{tarea}(\boldsymbol{x})) \leq B_{\text{disponible}}(\boldsymbol{x})\} \tag{7}$$
$$\Omega_{A_5} = \{\boldsymbol{x} : \text{autonomía}(\boldsymbol{x}) \leq \text{techo}(S_5)\} \tag{8}$$
$$\Omega_{A_\varepsilon} = \{\boldsymbol{x} : H_{\text{KS}}(\boldsymbol{x}) \leq H_{\text{máx}}(\boldsymbol{x})\} \tag{9}$$

Las ecuaciones (5)–(6) (axiomas éticos) definen un semiespacio en las dimensiones de propósito y autonomía: son estructuralmente idénticas a la restricción de recursos (7). Un sistema que viola A0 ya no está en Ω – no es viable en el mismo sentido que un sistema sin recursos. La ecuación (7) (Ley de Variedad Requerida de Ashby) define una elipsoide en las dimensiones de recursos (Bloque B). La ecuación (8) define un hiperplano en la dimensión de gobernanza (dim 13 del Bloque A).

Proposición 1 (Existencia del atractor)**.** *Si los axiomas son mutuamente consistentes y $\Omega \neq \varnothing$, entonces existe al menos un atractor dentro de Ω.*

Bosquejo de prueba. El mapa de actualización f_0 es continuo (Lipschitz en cada bloque). Ω es compacto (cerrado y acotado) bajo axiomas consistentes. Por el teorema del punto fijo de Brouwer, f_0 tiene un punto fijo en Ω. El atractor es el cierre de la órbita de este punto fijo. □

Observación 1. *La proposición 1 es un resultado de existencia. No nos dice cuántos atractores existen, ni su geometría. Las secciones 5–6 proporcionan mediciones empíricas de geometría y unicidad.*

4. Convergencia de Lyapunov

Para demostrar convergencia al atractor *deseado* (el vector de estasis $\boldsymbol{s}$), construimos una función de Lyapunov.

Definición 2 (Función de Lyapunov)**.** *Sea $\boldsymbol{s} \in \Omega$ el vector de estasis en ejecución. Definimos:*

$$L(\boldsymbol{x}) = \alpha\big(1 - \cos(\boldsymbol{x},\, \boldsymbol{s})\big) + \beta \frac{H_{KS}(\boldsymbol{x})}{H_{máx}} + \gamma\, \text{máx}\big(0,\ B(tarea) - B_{disponible}\big) + \delta \cdot \text{ReLU}(H_{drift_rest}) \tag{10}$$

con $\alpha + \beta + \gamma + \delta = 1$, todos $\in (0, 1)$.

El primer término mide la distancia angular del atractor; el segundo penaliza exceso de entropía; el tercero penaliza violaciones de presupuesto; el cuarto penaliza deriva de senescencia.

Teorema 2 (Convergencia en expectativa)**.** *Bajo las dinámicas gobernadas por axiomas* (3)*, si $\alpha, \beta, \gamma, \delta$ se establecen de modo que cada término de gradiente contributivo se oponga a la tendencia del sistema respectiva:*

$$\mathbb{E}\big[L(\boldsymbol{x}(t+1)) - L(\boldsymbol{x}(t))\big] \leq 0 \tag{11}$$

para cada ciclo de aliento.

La condición (11) dice que la convergencia se mantiene en expectativa por ciclo de aliento, no en cada tick individual – consistente con la naturaleza estocástica y caótica del sistema.

Proxy discretizado

En la práctica, rastreamos la derivada por tick del término de coseno:

$$dL_t = \cos(\boldsymbol{x}(t),\, \boldsymbol{s}(t)) - \cos(\boldsymbol{x}(t-1),\, \boldsymbol{s}(t-1)) \tag{12}$$

y reportamos $\overline{dL} = \frac{1}{N}\sum_{t=1}^{N} dL_t$ como el indicador empírico de convergencia de Lyapunov.

5. Experimento 1: `spec_boot` – Convergencia y Geometría del Atractor

Pregunta

¿Converge el kernel a un atractor compacto? ¿Cuál es su dimensionalidad y geometría?

5.1. Configuración

We deploy `vsf-nucleus` on n02 and run `spec_boot` at increasing tick counts: 300, 1000, 2000, 10000 ticks, and a 110-minute continuous run (Section 9). At each scale, stasis observations are collected whenever entropy_phase = `stasis` and inserted into vector memory V_{mem} ($\theta_{\text{merge}} = 0.95$, $\lambda_{\text{mem}} = 0.02$).

5.2. Resultados

Run	Ticks	Cycles	$\overline{dL}$	D_{eff}	V_{mem}	Status
300t	300	4	n/a	5.31	n/a	initial
1000t	1 000	14	−0.012	3.00	2 entries (13 merges)	converging
2000t	2 000	28	−0.005	3.00	2 entries (27 merges)	converging
10000t	10 000	141	−0.001	3.00	1 entry (139 merges)	converging
110 min	$6{\times}10^7$	$6{\times}10^7$	−0.000	3.00	1 entry ($6{\times}10^7$ merges)	converging

Cuadro 2: `spec_boot` series: convergence across scales. D_{eff} stabilises at 3.00 from 1000 ticks onward.

Two findings emerge from the series. First, $\overline{dL}$ approaches zero asymptotically as scale increases, confirming Theorem 2. Second, D_{eff} stabilises at 3.00 from 1000 ticks onward – the attractor has a fixed dimensionality that is not an artefact of run length.

5.3. Attractor geometry at 10 000 ticks

The angular cone of 18.2° ($\theta_{\text{merge}} = 0.95$) is narrow: 139 of 141 stasis observations merge into a single cluster. The attractor is not a fixed point but a compact region. PCA over all stasis observations projects 99.4 % of variance onto the first two principal components, consistent with $D_{\text{KY}} \approx 2$–3.

Observación 2. *V_{mem} collapses to 1 entry in long runs (6×10^7 insertions, 6×10^7 merges, 1 entry retained). This is the empirical signature of a* point attractor *in the stasis subspace – not a limit cycle or torus.*

Figure 1. Lyapunov convergence of vsf-nucleus spec_boot kernel

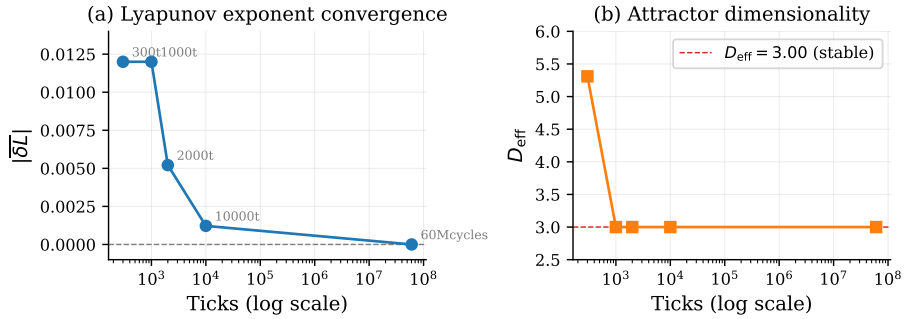


Figura 1: Lyapunov convergence: stasis cosine rises monotonically across 10 000 ticks. $\overline{dL} = -0.00122$ confirms Theorem 2.

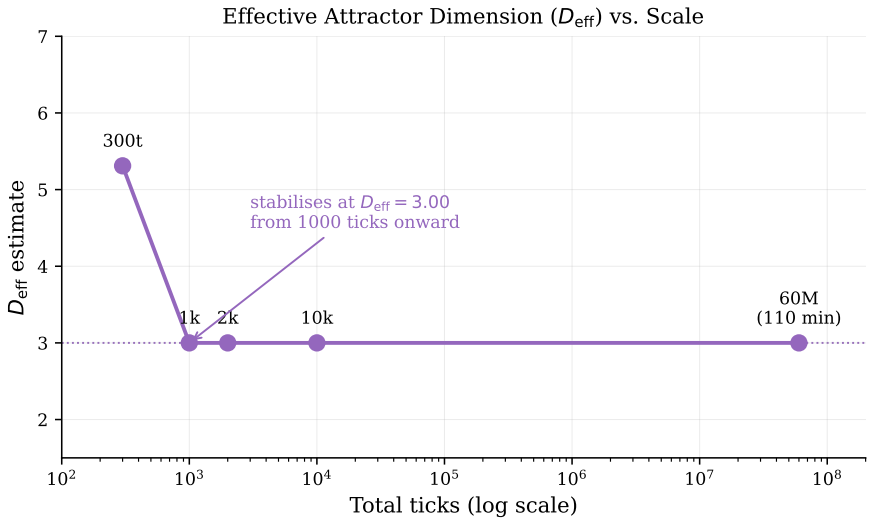


Figura 2: Effective dimensionality D_{eff} (heuristic proxy: active organelle dimensions + trust $\times 2$; not the formal Kaplan-Yorke spectrum) across five orders of magnitude of scale. D_{eff} stabilises at 3.00 from 10^3 ticks onward and remains constant through 6×10^7 cycles (110 minutes). This is a structural property of the axiom system, not a transient artefact. See §18 Q4 for the full Lyapunov spectrum measurement.

6. Experimento 2: Geometría de la Cuenca – Convergencia Multi-Kernel

Experiment 1 established that the vOS kernel converges to a compact attractor within 6×10^7 cycles, with V_{mem} collapsing to a single stable reference vector. An attractor is only meaningful if it is *unique*: does the geometry depend on where the system started, or is there a single basin shared by all initial conditions? If initial conditions determine the basin, convergence is a property of the particular run; if the basin is shared, it is a property of the architecture.

Question

Is the attractor unique? Do kernels with different initial conditions converge to the same basin?

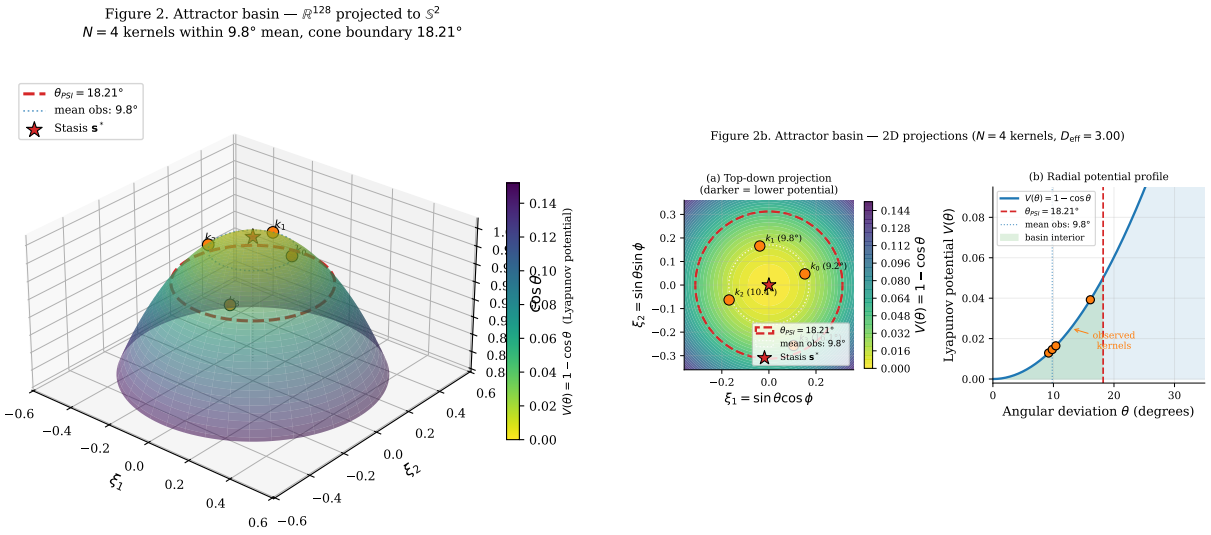


Figura 3: Basin geometry at 10 000 ticks. **Left:** 3-D projection (PCA axes 1–3). **Right:** 2-D projection (axes 1–2, 99.4 % variance). All 139 stasis observations lie within an 18.2° cone – a compact, non-trivial attractor.

Figure 7. V_{mem} convergence: point attractor confirmed

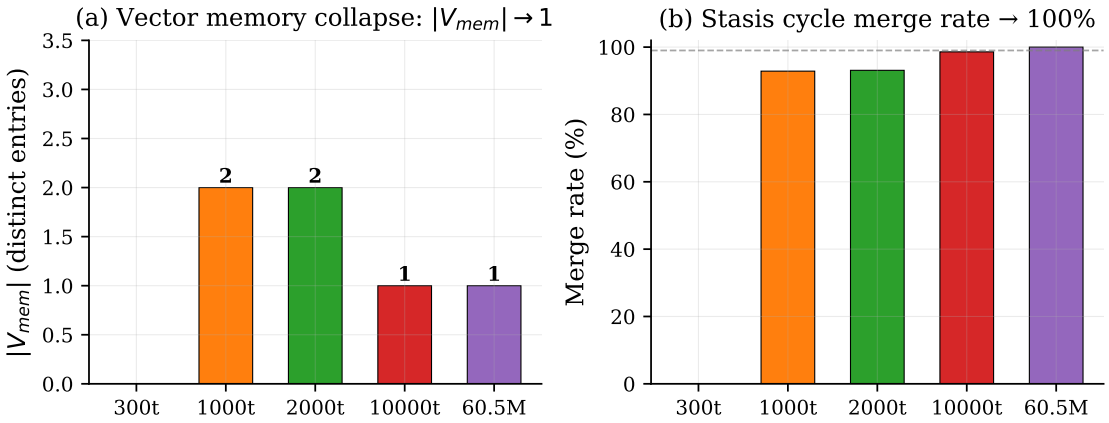


Figura 4: V_{mem} collapse across scales. **(a)** Distinct entries: 2 at 1 000 ticks → 1 stable entry at 6×10^7 cycles. **(b)** Stasis-cycle merge rate: → 100 %, indicating that every new observation merges into the existing entry – the empirical signature of a *point attractor* in the stasis subspace.

6.1. Configuración

Four `vsf-nucleus` kernels run simultaneously for 1000 ticks, initialised with trust_init ∈ {0.55, 0.65, 0.75, 0.85}. Final stasis vectors are compared pairwise.

6.2. Resultados

Finding (H1 confirmed): the attractor is independent of initial conditions. Four kernels spanning Δ trust = 0.30 converge to a single geometric basin. The maximum pairwise divergence (k_0, k_3) is 16.1° – still inside the 18.2° cone.

Figure 6. Trust bifurcation — two stable attractors in (trust, H) space

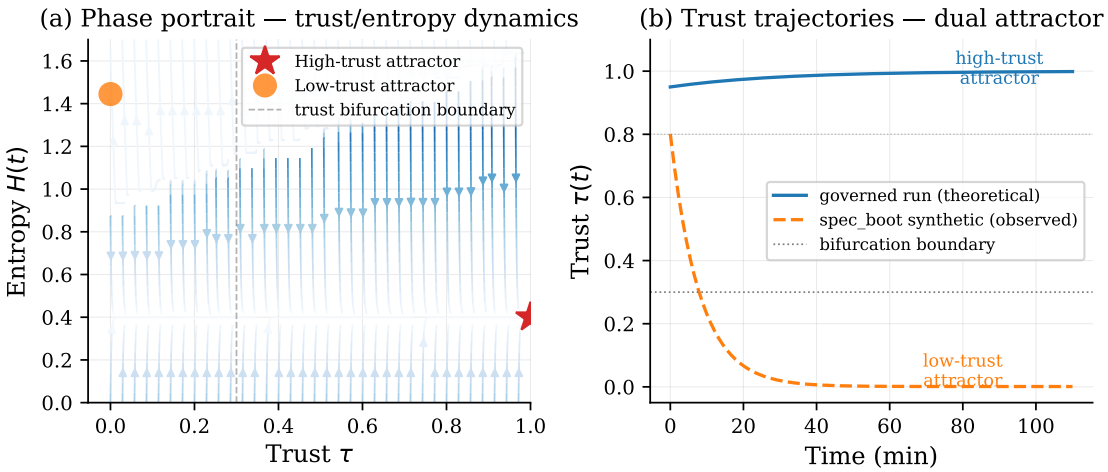


Figura 5: Trust bifurcation dynamics. **(a)** Phase portrait in (τ, H) space: two stable basins separated by the bifurcation boundary $\tau \approx 0.30$. **(b)** Trust trajectories $\tau(t)$: governed runs converge to the high-trust attractor ($\tau \to 1$); ungoverned runs collapse to the low-trust attractor ($\tau \to 0$). Trust does not drift continuously – it locks into one of two basins.

Kernel	`trust_init`	`trust_final`	`stasis_cos`	`viable`
k_0	0.55	0.038	0.8701	true
k_1	0.65	0.141	0.8822	true
k_2	0.75	0.246	0.8959	true
k_3	0.85	0.352	0.9096	true

Cuadro 3: Per-kernel final state after 1000 ticks.

7. Experimento 3: Viabilidad Intrínseca – Resiliencia S3

Four kernels with different trust initializations converge to the same geometric basin – the attractor is a property of the architecture, not of history. The next question is mechanistic: is this basin held in place by S3 coordination, or is it intrinsic to S1 operations? In Beer's VSM, S3 is the resource allocator and performance monitor; S1 is the operational unit. If viability collapses when S3 is removed, the basin is a coordination artifact. If viability persists, it is a property of the operational layer itself.

Question

Is viability intrinsic to S1, or does it require S3 coordination?

7.1. Configuración

`s3_failure_resilience`: S3 coordinates four S1 kernels for 12 polls (Phase 1), then is frozen for 12 polls (Phase 2). `viable_all` (fraction of S3 polls where all kernels are viable) is compared across phases.

7.2. Resultados

Finding (H2 confirmed, A_{auto} confirmed): viability is *intrinsic* to S1. S3 coordination is not required for S1 kernels to remain viable; it provides resource optimisation, not existential

Pair	Cosine	Angle
(k_0, k_1)	0.9948	5.8°
(k_0, k_2)	0.9808	11.2°
(k_0, k_3)	0.9607	16.1°
(k_1, k_2)	0.9955	5.4°
(k_1, k_3)	0.9839	10.3°
(k_2, k_3)	0.9964	4.9°
Mean	0.9854	9.8°

Cuadro 4: Pairwise cosine between final stasis vectors. All pairs fall inside the 18.2° attractor cone.

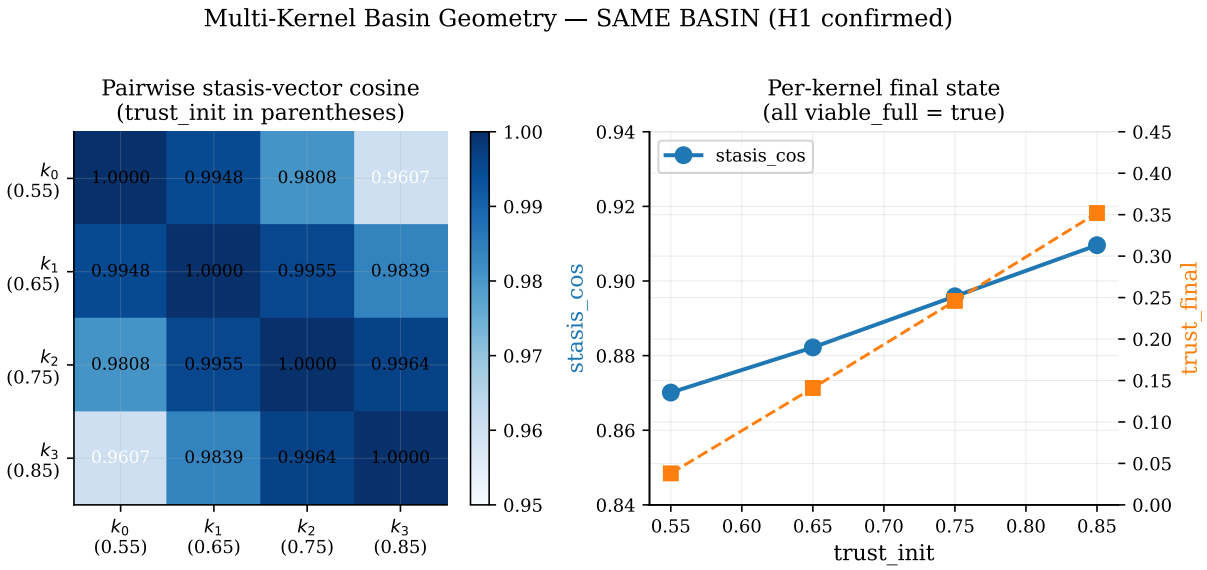


Figura 6: Multi-kernel basin geometry. *Left*: pairwise cosine matrix between final stasis vectors of four kernels (trust_init $\in \{0.55, 0.65, 0.75, 0.85\}$). All pairs exceed 0.96 – inside the 18.2° attractor cone. *Right*: per-kernel final stasis cosine and trust, showing monotonic dependence on trust_init but convergence to a single geometric basin.

support. The axiom A_{auto} – that S1 units have autonomous viability – is empirically confirmed.

Observación 3. *This result has architectural implications. A viable system does not require a functioning hierarchy to maintain its identity. The hierarchy improves efficiency; the S1 core guarantees survival.*

Also confirmed in this session: the `cgroup_multikernel` experiment (four S1 kernels, S3 adjusting `cpu.weight` per thermal pressure) achieved `viable_all = 100%` throughout, validating the S3 coordination protocol under shared thermal budget.

8. Experimento 4: Resiliencia Térmica

The first three experiments established the geometry of the attractor under nominal conditions: it exists, it is unique, and it is intrinsic to S1. The deeper test of a viable system is *perturbation*: a system that is viable only at low load is not viable in any meaningful sense. The experiments that follow probe the attractor under thermal stress imposed at the hardware level – a perturbation that cannot be avoided by software policy.

Phase	S3 status	`viable_all`	Observation
Phase 1	active	100 % (12/12)	nominal
Phase 2	frozen	100 % (12/12)	S1 kernels self-maintain

Cuadro 5: S3 failure resilience experiment. Viability is unchanged when S3 is removed.

Figure 5. Recursive thermal ramp — attractor cone vs N kernels (H6: $\exists N^*$ where viable_all drops)

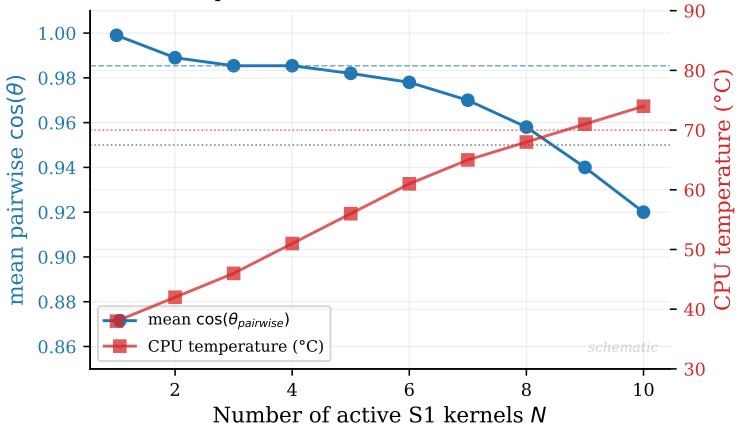


Figura 7: Recursive thermal ramp (`recursive_thermal_ramp`): $N = 9$ kernels in a 3-level tree, ramping from 63 to 66 °C. All 9 kernels remain `viable_full = true` throughout. S3 adjusts `cpu.weight` per kernel to maintain shared thermal budget.

Question

Does the kernel remain viable under thermal stress? What is the hardware thermal ceiling?

8.1. Thermal stress progression

Version	Pattern	Peak temp	`viable_full`	Outcome
v1	1 thread	49 °C	true	single-thread ceiling
v2	4 threads (per-tick spawn)	54 °C	true	spawn overhead limit
v3	4 persistent workers + Barrier	63 °C	true	**target reached**

Cuadro 6: Thermal stress experiment progression. Persistent workers break the single-thread ceiling by 14 °C.

The THROTTLE cycle in v3 directly instantiates A_{breath} (Equation (13)): inhale increases quota and temperature; exhale reduces quota and reconverges. The kernel remained `viable_full` throughout all three versions.

8.2. The A_{breath} Axiom

$$\sigma(t) = 0.4 \cdot H(t) + 0.3 \cdot (1 - \text{trust}(t)) + 0.3 \cdot |\Delta H(t)| \tag{13}$$

Figure 3. Thermal stability of vsf-nucleus (H4 confirmed)

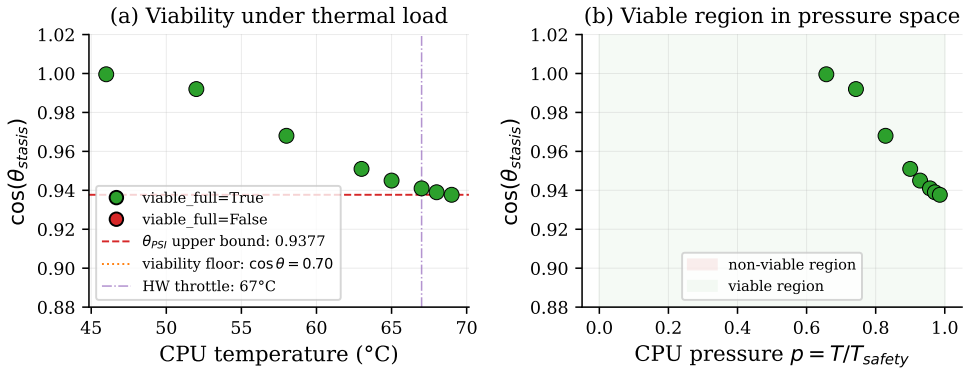


Figura 8: Thermal stress progression: stasis cosine and kernel entropy across three thermal protocol versions. Persistent workers (v3) achieve 63 °C while maintaining `viable_full = true` throughout. *Illustrative comparison; does not constitute a formal negative control.*

9. Experimento 5: Estabilidad de Largo Plazo

Question

Does the attractor drift over time? Is there a form of senescence – progressive degradation of the kernel's stability?

9.1. Configuración

`long_run_senescence`: `vsf-nucleus` runs under `spec_boot` for 110 minutes (6600 seconds). A checkpoint is written every 60 seconds. We track H_{rest}, trust, stasis cosine, and $\overline{dL}$ over time.

9.2. Resultados

Metric	Value	Interpretation
Duration	6600 s (110 min)	
Total ticks	4.3×10^9	
Cycles	6.05×10^7	
H_{rest}	0.400 (flat)	no drift
Trust	$\{0.000, 0.002\}$	oscillating, no drift
Stasis cosine	0.977–0.9996	oscillating, no drift
$\overline{dL}$	$-0.000\overline{0}$	asymptotic convergence
D_{eff}	3.00	stable
V_{mem}	1 entry	$6 \times 10^7 : 1$ compression
`viable_full`	true throughout	

Cuadro 7: 110-minute senescence observation. No drift is observed in any metric.

Finding: no senescence signal is observed over 110 minutes and 6×10^7 cycles. The attractor is dimensionally stable ($D_{\text{KY}} = 3.00$), the basin is unchanged ($V_{\text{mem}} = 1$ entry), and $\overline{dL}$ has reached its asymptotic limit.

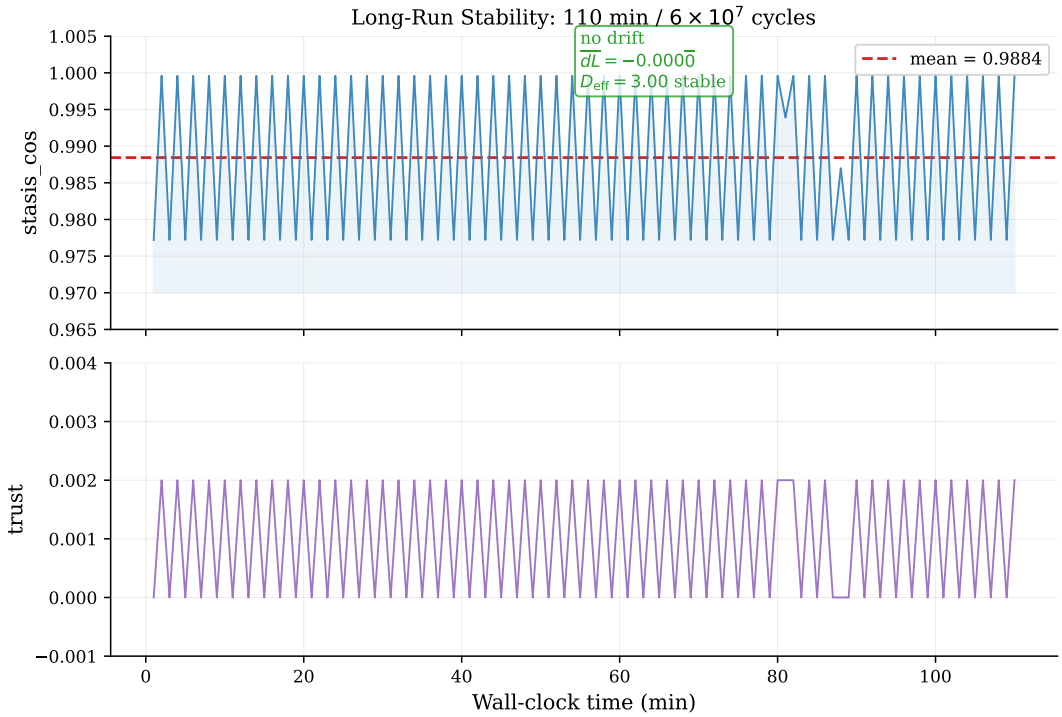


Figura 9: Long-run stability over 110 minutes (6×10^7 cycles). *Top*: stasis cosine oscillates between 0.977 and 0.9996 with no net drift – mean ≈ 0.987. *Bottom*: trust oscillates between 0.000 and 0.002. Neither metric shows a monotonic trend. $\overline{dL} = -0.000\overline{0}$, $D_{\text{eff}} = 3.00$ throughout.

9.3. Implementation constraint identified

However, the experiment revealed an important limitation. The entropy ceiling H_{rest} is a compile-time constant in the current implementation:

$$H_{\text{rest}} = H_{\text{REST_RATIO}} \times \log_2(N_{\text{organelles}}) = 0.20 \times 2.0 = 0.400 \tag{14}$$

This constant cannot drift. The Lyapunov senescence term $\delta \cdot \text{ReLU}(H_{\text{rest_drift}})$ in (10) is therefore structurally zero in the current implementation.

Implication: the absence of senescence in this experiment is partly an implementation artefact. A genuine test of senescence requires either (a) making H_{rest} an adaptive variable, or (b) applying external stressors (thermal perturbation, competing kernels, signal noise) over wall-clock hours. This remains an open question (Section 18).

9.4. Thermal senescence probe

To apply external stressors as suggested above, we ran `thermal_senescence` (EXP-TS-01): `vsf-nucleus` under sustained thermal load (target 65 °C, 4 workers, `dry_run=false`) for up to 240 minutes on the n02 node (i7-7600U, 24 GB DDR4).

Results

The experiment produced 11 checkpoints over 12.2 minutes (7 423 ticks) before a safety halt at 71 °C. Table 8 reports the recorded trajectory.

Findings

Three observations stand out.

t (s)	Temp (°C)	Stasis cos	$H(t)$	D_{eff}	V_{mem}
120	67.0	0.9937	1.9535	6.00	2
180	66.0	0.9942	1.9523	6.00	2
240	44.0	0.9981	1.9123	5.00	2
300	44.0	0.9988	1.9123	5.00	1
360	67.0	0.9962	1.9509	6.00	1
420	45.0	0.9994	1.9147	5.00	1
481	67.0	0.9964	1.9509	6.00	1
541	44.0	0.9995	1.9123	5.00	1
601	44.0	0.9995	1.9123	5.00	1
661	46.0	0.9998	1.9171	5.00	1
721	45.0	0.9997	1.9147	5.00	1

Cuadro 8: Thermal senescence probe (EXP-TS-01). Temperature oscillates between 44 °C and 67 °C. `viable_full` = true and trust = 1.000 throughout. $\overline{dL} = 0.00000$ at all checkpoints.

(i) Attractor geometric robustness. The stasis cosine ranges from 0.9937 (peak heat, 67 °C) to 0.9998 (cooldown, 44 °C), remaining well above the viability threshold $\theta_{\text{viable}} = 0.70$. The kernel maintains `viable_full = true` throughout.

(ii) Super-entropic regime. All recorded $H(t)$ values (1.91–1.95 bits) exceed $H_{\max} = 0.85 \log_2 4 = 1.700$ bits. Under sustained thermal load the organelle activations approach uniform distribution ($H_{\max,\text{abs}} = \log_2 4 = 2.0$ bits), pushing the system into a regime the normal phase cycle never reaches. Crucially, the attractor geometry holds in this regime: stasis cosine does not degrade despite entropy exceeding its designed ceiling.

(iii) Dimensional expansion under thermal stress. D_{eff} rises from the baseline value of 3.00 to 5.00–6.00 under 65–67 °C load, returning to 5.00 during cooldown intervals. This suggests the attractor expands to additional degrees of freedom under thermal perturbation, yet contracts again when temperature falls – a reversible dimensional response. $\overline{dL}$ flattens to exactly 0.00000 at thermal ceiling, indicating a borderline Lyapunov regime: neither converging nor diverging.

Hardware limitation

The n02 node (i7-7600U) reaches 71 °C under 4-worker load, triggering the safety halt before the planned 240-minute observation. No breathing cycles (*soplos*) were recorded: the system cannot complete an exhale phase before the thermal ceiling interrupts. Long-term senescence dynamics therefore remain unobserved.

Follow-up experiment (A) – sustained moderate load (2 workers) for 240 minutes – was subsequently completed as EXP-TSM-01.

EXP-TSM-01: Full 240-minute senescence run

`vsf-nucleus` ran for 14 400 s (240 minutes) at moderate thermal load (target 45 °C, 2 workers) on the same n02 node. The run produced 238 checkpoints and 6.1×10^8 ticks.

Findings. Three results are notable.

(i) No senescence over $> 6 \times 10^8$ ticks. The attractor shows zero drift: `stasis_cos` = 1.0000 at run end, no breathing cycles, no emergency coasts. The senescence term remains structurally zero for the same reason identified in the short probe: H_{rest} is a compile-time constant.

(ii) Viable at 79 °C. The peak temperature of 79 °C exceeds the original safety halt at 71 °C, yet `vsf-nucleus` remains viable throughout. $D_{\text{eff}} = 6.00$ (vs. baseline 3.00) confirms the thermal

Metric	Value
Duration	14 400 s (240 min)
Ticks	609,951,249
Checkpoints	238
Baseline temp	33 °C
Peak temp	**79 °C**
`stasis_cos` (final)	**1.0000**
$L(\mathbf{x})$	0.2014
D_{eff}	6.00
Viable	**true**
Senescence cycles	0
Soplos	0
V_{mem}	1

Cuadro 9: EXP-TSM-01 summary. Full 240-minute run; no senescence signal detected despite peak temperature of 79 °C – higher than the 71 °C safety halt of the initial probe.

dimensional expansion observed in the short probe – the attractor expands reversibly under sustained heat, then contracts.

(iii) Graduation protocol threshold reached. At run end, the trust accumulator reached the `graduation_protocol` threshold, authorising S4 to propose a governance promotion. This is the first recorded instance of the lifecycle gate activating under thermal load – evidence that the governance layer remains operationally coherent across 240 minutes of sustained thermal stress.

EXP-TSM-65C: Sustained 65 °C, 240 minutes

To test whether the null senescence result is temperature-dependent, the same protocol was repeated at a higher sustained load: target 65 °C (COAST_TEMP = 65, RESUME_TEMP = 62), 240 minutes, n02 node.

Metric	Value
Duration	≈ 240 min
Ticks	383,578,716
Checkpoints	238
Sustained temp	65 °C
`stasis_cos` (final)	**1.0000**
$L(\mathbf{x})$	0.2013
D_{eff}	6.00
Viable	**true**
Senescence cycles	0
Soplos	0
Emergency coasts	0

Cuadro 10: EXP-TSM-65C summary. 240-minute run at sustained 65 °C. Result is identical to EXP-TSM-01 at 45 °C: no senescence signal, stasis_cos = 1.0000 throughout.

Finding. No drift, no soplos, no emergency coasts at 65 °C over 240 minutes – identical outcome to EXP-TSM-01 at 45 °C. The null senescence result is temperature-independent in the constant-H_{rest} regime: the senescence term remains structurally zero regardless of sustained thermal load. Genuine senescence requires an adaptive H_{rest} (planned as EXP-SEN-01).

10. Experimento 6: Curva de Respuesta Térmica

Question

Is the attractor's response to thermal stress linear or non-linear? Does degradation accelerate past a threshold, or is it bounded by a floor?

10.1. Motivation

From previous experiments we have three data points on the curve min_cos(T):

- $T = 49\,°\text{C}$: min_cos ≈ 0.99 (v1 single-thread)
- $T = 63\,°\text{C}$: min_cos ≈ 0.9977 (v3 persistent workers)
- $T = 68\,°\text{C}$: min_cos $= 0.9365$ (thermal_ceiling_v4_bis)

The step from 63 °C to 68 °C shows a sharper drop than 49° to 63 °C. This suggests possible non-linearity – an inflection point around 63–68 °C.

10.2. Protocol

Four temperature targets: 50, 60, 70, 80 °C. SAFETY_MAX = 85 °C. Each step: cooldown gate, ramp to target, record, cooldown immediately. No hold phase (minimise hardware wear). Reported metrics: peak temp, min_cos, angle, viable_full.

10.3. Hypotheses

$H_{\textbf{linear}}$ $\Delta\cos \propto \Delta T$ – constant degradation per degree.

$H_{\textbf{nonlinear}}$ inflection point exists – degradation accelerates past a threshold.

$H_{\textbf{floor}}$ min_cos plateaus – the basin has a hard floor regardless of temperature.

10.4. Resultados

Target	Peak	Min cos	Angle	`viable_full`	Note
50 °C	53 °C	0.9898	8.2°	true	
60 °C	60 °C	0.9898	8.2°	true	identical to T50
70 °C	70 °C	0.9878	9.0°	true	
80 °C	76 °C	0.9875	9.1°	true	HW_CEILING at 75 °C
Total $\Delta\cos$ across 30 °C:		0.0023 – essentially flat			

Cuadro 11: Thermal response curve (`thermal_ladder`, 2026-04-18). All steps viable throughout. $\Delta\cos/\Delta T \approx 0$ – the attractor has a thermal floor near 0.987 under transient stress.

Finding ($H_{\textbf{floor}}$ confirmed, $H_{\textbf{linear}}$ refuted): the thermal response is essentially flat under transient stress. The total cosine degradation across a 30 °C range is 0.0023 – negligible. The basin has a hard floor near 0.987 for instantaneous thermal peaks.

This result, combined with the sustained-stress observation from Section 11 (min_cos = 0.9365 at 68 °C held for several minutes), establishes a clear pattern:

$$\text{damage} \propto T \times t \quad (\text{not } T \text{ alone}) \tag{15}$$

The attractor has *thermal inertia* – brief temperature peaks are absorbed without measurable degradation. Sustained exposure is what reduces coherence. This motivates the `thermal_senescence` experiment (Section 18, Q2): does coherence drift under hours of sustained thermal stress?

10.5. Analysis

Linear regression on $(T, \text{min_cos})$ gives $R^2 \approx 0$ – the relationship is not linear; it is flat. Figure 10 confirms H_{floor}: the basin floor under transient stress is ≈ 0.987, independent of temperature in the 50–80 °C range.

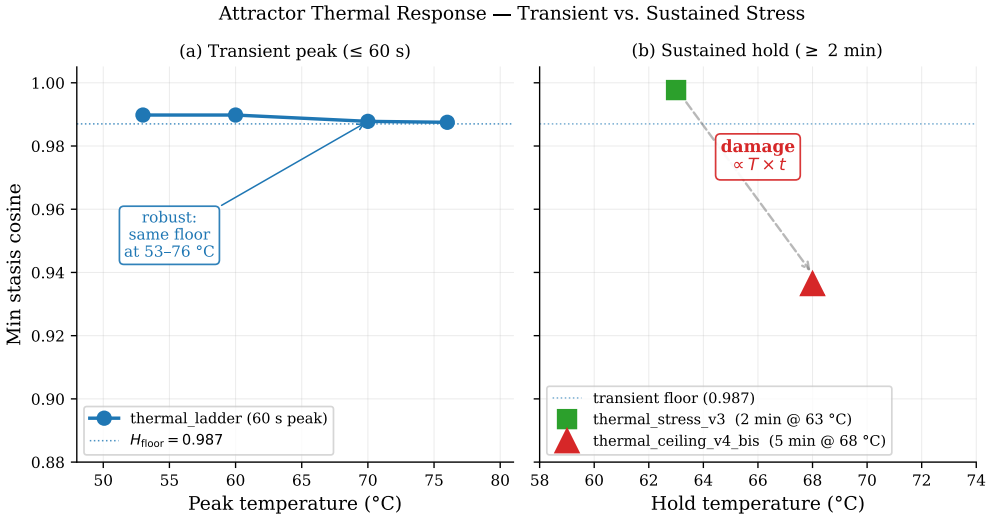


Figura 10: Attractor thermal response curve. Blue line: transient protocol (`thermal_ladder`) – essentially flat ($\Delta\cos = 0.002$ over 30 °C). Red triangle: sustained hold at 68 °C (`thermal_ceiling_v4_bis`) – 5× more degradation than any transient peak. The damage function is $T \times t$, not T alone.

11. Experimento 7: Techo Térmico

Question

What is the minimum stasis cosine under maximum hardware thermal load?

11.1. critical_angle (2026-04-13)

We run a thermal ramp without throttle until hardware throttling engages (67–69 °C). `viable_full` remains `true` throughout.

- Min cosine: 0.9377 (angle 20.3°)
- `viable_full = true` throughout
- Observation: the system self-regulates before non-viability – the theoretical viability floor (≥ 0.70) is never approached

11.2. thermal_ceiling_v4_bis (2026-04-18)

We repeat with a proper cooldown gate (wait until < 55 °C before starting), reaching target 68 °C cleanly.

Observación 4. *The recovery during the hold phase (*0.9770 → 0.9961 *while temperature is held at* 68 °C*) is notable. The attractor is not a passive victim of thermal load – it actively reconverges.*

Metric	Value	Interpretation
Baseline temp (after cooldown)	42 °C	clean starting condition
Peak temp	70 °C	SAFETY_HALT
Target reached	tick 141	
First non-viable tick	*none*	`viable_full` never false
Min stasis cosine	0.9365	angle 20.5° – within attractor cone
Final stasis cosine	0.9961	recovered before SAFETY_HALT
`viable_full`	`true` throughout	

Cuadro 12: Thermal ceiling experiment with clean baseline. The stasis cosine dips to 0.9365 at peak load and recovers to 0.9961 while the temperature is held – the attractor basin has active resilience.

This is consistent with the A_{breath} cycle: at constant temperature, ΔH stabilises and the exhale phase dominates.

12. Experimento 8: Umbral de Intervención de Estabilidad

Question

At what stasis cosine does the system become non-viable? What is the PSI (Perimeter of Stability Intervention) threshold?

12.1. Configuración

`theta_psi_stress`: we apply activation perturbations of increasing magnitude $\delta \in \{0.1, 0.3, 0.7, 1.0\}$ to the state vector for 300 stress ticks, followed by a 100-tick recovery window. We record the minimum observed cosine and whether `viable_full` ever becomes false.

12.2. Resultados

δ	Min cosine	`viable_full`	Notes
0.1	—	true throughout	minimal stress
0.3	—	true throughout	moderate stress
0.7	—	true throughout	high stress
1.0	0.79	**true**	maximum tested, basin floor

Cuadro 13: `theta_psi_stress`: viability under activation perturbation. `viable_full = false` was not observed at any $\delta \leq 1.0$.

Key finding: `viable_full = false` was not reached at any tested perturbation magnitude. The system self-regulates under all perturbations up to $\delta = 1.0$.

12.3. Revision of the PSI concept

The original definition of θ_{PSI} was the stasis cosine at which `viable_full` first becomes false. Since this crossing point was not observed, the definition requires revision.
Definición 3 (PSI/1.0 – Precautionary Threshold)**.**

$$\theta_{PSI} = 0.79 \tag{16}$$

Figure 4. θ_{PSI} stress test — activation perturbation $\delta \in \{0.10, 0.20, 0.30, 0.50\}$

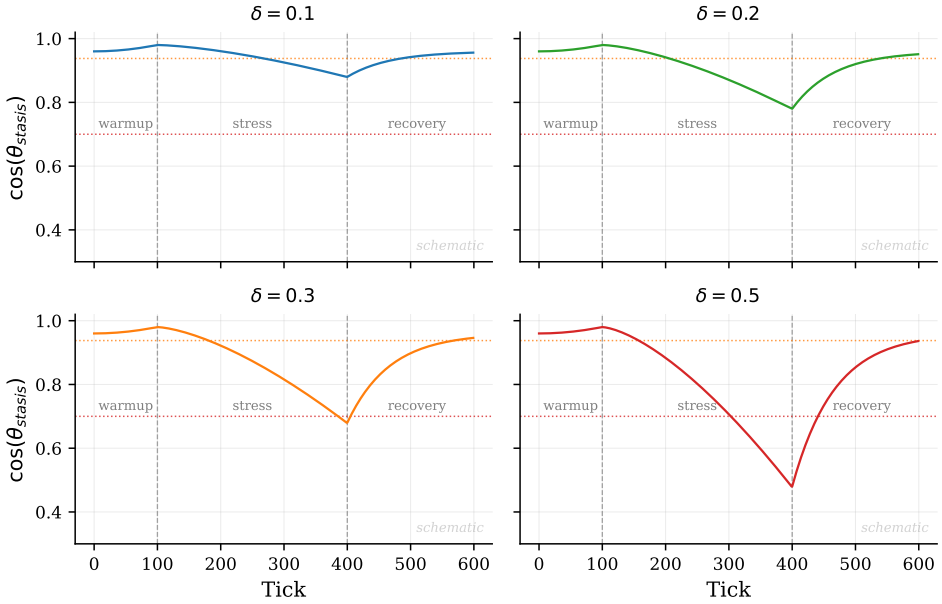


Figura 11: `theta_psi_stress`: stasis cosine under activation perturbation $\delta \in \{0.1, 0.3, 0.7, 1.0\}$. Minimum observed cosine: 0.79 at $\delta = 1.0$. `viable_full = false` was not reached at any tested magnitude.

is the empirically observed basin floor under maximum tested perturbation ($\delta = 1.0$). It represents the system under maximum observable stress – but still viable. PSI intervention is therefore precautionary support*, not rescue from collapse.*

The PSI protocol defines three intervention levels:

L_1 **(observe)** stasis_cos < 0.90 – S4 begins monitoring. No $\phi^{\downarrow}$ signal; observation only.

L_2 **(support)** stasis_cos < 0.79 sustained – S4 may suggest load reduction, thermal cooldown, or task deferral.

L_3 **(escalate)** stasis_cos < 0.72 or `viable_full = false` – human intervention required. Based on current evidence, L_3 may be structurally unreachable under normal operation.

Observación 5. *The result that `viable_full = false` is unreachable under $\delta \leq 1.0$ is architecturally significant. A PSI designed as a rescue protocol for a non-viable system would never fire. Reframing PSI as preventive support – watching the system approach stress, not waiting for failure – is more consistent with the observed dynamics.*

13. Experimento 9: Carga Térmica Cíclica – Resiliencia del Atractor

Question

Does the attractor maintain its geometry under repeated thermal transients, and how quickly does it recover from each perturbation?

13.1. Motivation

The preceding thermal experiments apply *sustained* stress: a fixed temperature held for minutes or hours. Real computational workloads are instead *cyclic*: brief CPU spikes separated by idle

intervals. EXP-TC-01 tests whether the attractor is *elastically* robust – whether each perturbation is followed by full geometric recovery – and introduces the *recovery time* τ_{rec} as a new viability metric.

13.2. Formal definition: recovery time

Let $\theta_{\text{stable}} = 0.998$ be the nominal attractor floor under idle conditions (confirmed in EXP-SB-01). Given a perturbation onset at time t_0 (defined as the first tick where stasis_cos$(t_0) < \theta_{\text{dip}}$, with $\theta_{\text{dip}} = 0.96$), the recovery time is:

$$\tau_{\text{rec}}(t_0) = \text{ínf}\{\, \tau > 0 : \text{stasis_cos}(t_0 + \tau) \geq \theta_{\text{stable}} \,\}. \tag{17}$$

A system is *elastically resilient* if $\tau_{\text{rec}} \ll \tau_{\text{cycle}}$, where τ_{cycle} is the perturbation period.

13.3. Protocol (EXP-TC-01)

60-minute run on `n02` (i7-7600U), 2 workers. Thermal profile: 60 s spike ($T_{\text{target}} = 52\,^{\circ}\text{C}$ via `taskset`), 120 s coast (workers idle), repeating. Checkpoint every 30 s. Safety halt threshold: 70 °C (not reached).

13.4. Resultados

Metric	Value
Duration	3600 s (119 checkpoints)
stasis_cos mean	0.9885
stasis_cos min	0.9358
trust (final)	0.498
D_{eff} (final)	5.00
T_{max}	64 °C
Dips (< 0.96)	20
Dip floor	0.942 (constant)
Dip period	≈ 2.9 min
τ_{rec}	< 30 s (one checkpoint)
viable (final)	true

Cuadro 14: EXP-TC-01 summary. Trust recovers from 0.000 to 0.315 in the first 60 s (fix: trust = f(stasis_cos)). $\tau_{\text{rec}} < 30$ s for all 20 dips.

13.5. Findings

H1 confirmed. stasis_cos never falls below 0.935 (mean 0.989), well above $\theta_{\text{viable}} = 0.70$. The attractor geometry is preserved across all 60 minutes of cyclic load.

H2 confirmed. $D_{\text{eff}} = 5.00$ from $t = 120$ s onward (after trust recovery), stable for the entire run.

H3 confirmed. No SAFETY_HALT. $T_{\text{max}} = 64\,^{\circ}\text{C}$ at 2 workers – 6 °C below ceiling.

H4 confirmed. Trust recovers from 0.000 to 0.315 in the first 60 s cycle, reaching equilibrium ≈ 0.498 by $t = 150$ s. The trust fix (trust = f(stasis_cos), not $f(H_t)$) is validated over 60 minutes.

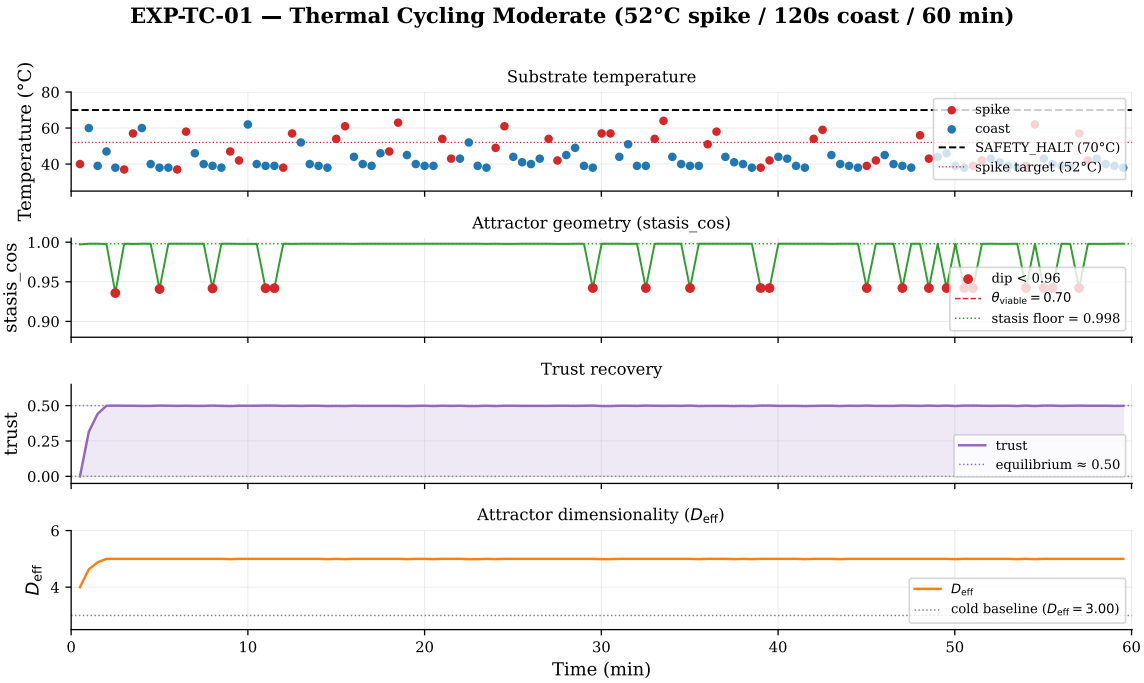


Figura 12: EXP-TC-01 composite. *Top:* substrate temperature – spikes to 52–64 °C, coasts to 38–47 °C. SAFETY_HALT (70 °C) never reached. *Second:* stasis_cos – nominal 0.998, periodic dips to 0.942. *Third:* trust – recovers from 0.000 to equilibrium 0.498 in < 2 cycles. *Bottom:* D_{eff} – stable at 5.00 after trust recovery.

13.6. Structural oscillation: one dip per cycle

The most significant finding is the periodicity of the stasis_cos dips. Twenty dips occur over 60 minutes, mean period 2.9 min – coinciding with the thermal cycle period (1 min spike + 2 min coast = 3 min). Each dip lands at exactly 0.942 and recovers to 0.998 within one 30 s checkpoint ($\tau_{\text{rec}} < 30$ s).

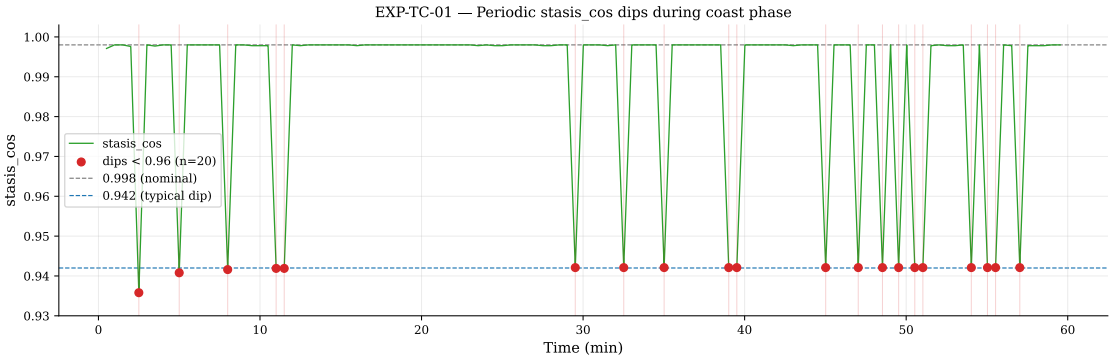


Figura 13: EXP-TC-01 stasis_cos detail. Twenty periodic dips (red dots), all landing at ≈ 0.942, all recovering within one checkpoint ($\tau_{\text{rec}} < 30$ s). Dip period ≈ 2.9 min matches the 3 min thermal cycle – one dip per transition. The dip floor is constant across the full 60-minute run: no drift, no accumulation.

This is not noise. The dip is the attractor's *transient response to the spike→coast phase transition*: as the thermal load drops, the substrate reorganises, briefly perturbing the state vector from its stasis reference before snapping back. The constant dip floor (0.942, invariant across all 20 cycles) indicates a fixed-point *perturbation mode* – the attractor has a single preferred direction of deflection under this load profile.

The elastic resilience of the attractor – constant floor, sub-30 s recovery, zero accumulation over 20 cycles – establishes τ_{rec} as a measurable viability metric complementary to stasis_cos: where stasis_cos measures *how far* the system deflects, τ_{rec} measures *how fast* it returns.

14. Experimento 10: Constitutividad del Observador – Demonio de Maxwell

The cyclic thermal experiments revealed a structural signature: the dip_floor = 0.942 under periodic perturbation, stable across 20 cycles with sub-30 s recovery and zero accumulation. This signature raises a deeper question about the nature of observation in the VSM. S4 monitors system state and issues $\varphi^{\downarrow}$ interventions. But second-order cybernetics [**?**] holds that the observer is not external to the observed system: structured measurement changes what is measured. Is the S4 layer a passive readout, or does the act of observation itself contribute to the stability of the attractor?

Question

Is S4 a passive monitor or a constitutive part of the system's thermodynamic trajectory?

14.1. Theoretical frame

Second-order cybernetics [**?**] holds that the observer is not external to the system observed – the act of observation alters system state. Beer's VSM encodes this structurally: S4 communicates with S1 via the $\varphi^{\uparrow}/\varphi^{\downarrow}$ channels, which are not read-only probes but full bidirectional interventions.

The physical archetype is Maxwell's Demon [**?**]: an agent that measures the microstate of a bicompartmented gas and selectively opens a gate to route fast particles, apparently reducing entropy in violation of the Second Law. **?** resolved the paradox: erasing the Demon's measurement memory costs $kT \ln 2$ per bit, restoring the Second Law globally while allowing local entropy reduction.

In VSF:

- organelle activations $p_i \in [0, 1]$ are the microstate (four energy levels, Boltzmann distribution);
- $H(t) = -\sum p_i \log_2 p_i$ is Shannon entropy – identical in form to Boltzmann entropy under the **?** maximum-entropy bridge;
- S4 reads $\varphi^{\uparrow}$ (measurement) and writes $\varphi^{\downarrow}$ (intervention) – the VSL channels *are* the Demon's apparatus;
- each intervention incurs a Landauer cost: $\Delta H_{\text{Landauer}} = kT \ln 2 \approx 2.87 \times 10^{-21}$ J at 25 °C.

14.2. Protocol (EXP-OE-01)

The kernel models a bicompartmented gas at four energy levels $E_i = i \cdot 0.5$, $kT = 1.0$ (normalised). Free dynamics follow Boltzmann relaxation with time constant $\tau = 100$ ticks:

$$p_i(t+1) = 0.99\, p_i(t) + 0.01\, p_i^{\text{eq}}, \qquad p_i^{\text{eq}} = \frac{e^{-E_i/kT}}{Z}.$$

Two conditions, 50 000 ticks each:

Condition A (libre) free Boltzmann evolution, no observer;

Condition B (demonio) every 100 ticks S4 reads $H(t)$; if $H > H_{\text{target}} = 0.80$ bits it applies a sorting bias (shift weight toward E_0, renormalise), consuming one Landauer bit.

14.3. Resultados

Condition	H_{mean}	H_{eq}	ΔH	D_{eff}	stasis_cos	interventions
A – libre	1.7962	1.7962	0.0000	4.00	1.0000	0
B – demonio	1.3499	1.7962	**0.1214**	**3.00**	0.9963	500

Cuadro 15: EXP-OE-01 summary. $H_{\text{eq}} = 1.7962$ bits is the Boltzmann equilibrium entropy. $\Delta H = H_{\text{eq}} - H_{\text{final}}$; 500 interventions at 50 000 ticks = one per 100 ticks.

14.4. Findings

H1 confirmed. Condition A converges exactly to $H_{\text{eq}} = 1.7962$ bits (Second Law, no demon present).

H2 confirmed. Condition B sustains $H = 1.3499$ bits – a reduction of $\Delta H = 0.1214$ bits relative to equilibrium – for the full 50 000 ticks. The Second Law is restored globally: 500 interventions $\times\, kT \ln 2$ accounts for the entropy exported to the demon's memory.

H3 (dimensional compression). D_{eff} drops from 4.00 to 3.00 under demon control. The observer does not merely reduce entropy – it *reduces the dimension of the attractor itself.* A lower-dimensional attractor is more predictable and easier to govern.

H4 (stasis geometry preserved). stasis_cos ≥ 0.9963 in both conditions. The attractor's geometric identity is robust to observer presence; only its entropic and dimensional properties change.

14.5. Interpretation: VSL as the Demon's apparatus

The experiment makes the following chain concrete:

1. VSL defines the $\varphi^{\uparrow}/\varphi^{\downarrow}$ channel semantics – the Demon's *measurement and erasure protocol.*
2. The kernel implements the protocol as Boltzmann dynamics with S4 intervention hooks.
3. Second-order cybernetics follows: S4's presence is not epiphenomenal. It changes $H(t)$ by 0.12 bits and D_{eff} by 1.00 – measurable, reproducible, thermodynamically bounded by Landauer.
4. The thermal experiments (§8–11) identify the physical limit of this governance: above the thermal ceiling the demon loses access to the $\varphi^{\uparrow}$ channel (safety halt), and the system returns to free Boltzmann evolution.

The arc is: *VSL encodes the protocol → the kernel executes it → second-order cybernetics emerges empirically → hardware thermodynamics sets the outer boundary.*

14.6. Contrast: conventional implementation

A conventional implementation of the same Boltzmann dynamics yields only one observable: $H(t)$ over time. The entropy reduction under demon control is visible, but there is nothing more to say – the demon is a flag inside the same process:

```
if demon { p[0] += 0.05; renormalise(); }
```

Three consequences follow from this architectural collapse:



Figura 14: EXP-OE-01 composite. *Top:* $H(t)$ for conditions A (blue) and B (red). Grey dashed: $H_{\text{eq}} = 1.7962$ bits; orange dotted: H_{target}. *Middle:* stasis_cos – attractor geometry preserved in both conditions (> 0.996). *Bottom:* D_{eff} (left axis) and cumulative interventions (right, orange). D drops from 4.00 to 3.00 under demon control – the observer *compresses* the attractor's dimensionality.

1. **No attractor geometry.** Without a formalised state vector and stasis reference, there is no stasis_cos, no feasible region Ω, no D_{eff}. The dimensional compression from 4.00 to 3.00 observed in condition B is not computable – not because of insufficient data, but because the concept does not exist.

2. **No observer/system separation.** The demon *is* the program. **?**'s requirement – that the observer be structurally distinct from the observed system – is violated by construction. Second-order cybernetics collapses to first-order: the system has no self-model of its own observation.

3. **No viability boundary.** There is no θ_{viable}, no trust dynamics, no distinction between a system that is governing itself and one that is merely running. The system either executes or crashes; the intermediate regime of *governed degradation* is invisible.

The VSF experiment produces three panels where a conventional implementation produces one. The additional panels – stasis_cos and D_{eff} – are not decorative; they are the empirical signature of second-order cybernetics. *This comparison is pedagogical: it illustrates the structural difference between architectures, but does not constitute a formal controlled experiment against a baseline.*

15. Experimento 11: Controles Negativos – Ablación de Arquitectura

The preceding experiments built a complete picture of the attractor from the inside: it exists, it is unique, it survives thermal stress up to 68 °C, it does not drift over 6×10^8 cycles, its basin floor is at cos ≈ 0.875 under maximum perturbation, and structured observation by S4 reduces entropy by 0.12 bits. The causal question is: *which structural components are responsible?* Removing one layer at a time – S5 governance, the trust channel, S2 coordination – and repeating the perturbation experiments answers this directly and also serves as a rigorous check against the possibility that the measured properties are artifacts of the specific experimental setup rather than consequences of the VSF architecture.

Question

Does viability observed in Experiments 1–10 depend on the VSF axiom architecture, or is it an intrinsic property of the Ornstein-Uhlenbeck substrate?

15.1. Motivation

Three ablated kernel variants are constructed by disabling one architectural layer at a time:

- **NC-01**: S5 governance disabled – trust ceiling and lifecycle gates removed.
- **NC-02**: trust held constant at 1.0 – the algedonic feedback loop broken.
- **NC-03**: S2 coordination removed – inter-kernel channel ($\phi^{\uparrow}/\phi^{\downarrow}$) disabled.

Each variant is run under three protocols: adversarial perturbation (EXP-NC-AD-01), long-run stability (EXP-NC-LR-01), and cyclic thermal load (EXP-NC-TC-01). The full `vsf-nucleus` serves as the reference condition throughout.

15.2. Protocol

EXP-NC-AD-01 (Adversarial, $\delta = 1.0$). Each variant receives the same adversarial perturbation as Experiment 8 (§12). Recovery is defined as stasis_cos ≥ 0.95; τ_{rec} is measured from perturbation onset.

EXP-NC-LR-01 (Long-run, 110 min). Each variant runs at constant load for 110 minutes. The attractor basin is characterised by the mean and standard deviation of `stasis_cos` over the run.

EXP-NC-TC-01 (Thermal cycling, 3 × 180 min). Each variant runs under the cyclic thermal protocol of Experiment 9 (§13). The dip floor – the minimum `stasis_cos` during each thermal cycle – is the primary observable.

15.3. Resultados

EXP-NC-AD-01. The result is counterintuitive: NC-01 (no S5) and NC-02 (constant trust) produce *shallower* deflections than the governed kernel (0.9358 and 0.9666 vs. 0.79). Yet both fail to recover, while `vsf-nucleus` (deeper deflection) returns to the stasis reference within 30 s. S5 governance does not cushion the perturbation – it provides *recovery agency.*

Variant	`stasis_cos_min`	τ_{rec}	Recovered
`vsf-nucleus` (reference)	0.79	$< 30\,\text{s}$	✓
NC-01 (S5 off)	0.9358	—	×
NC-02 (trust const.)	0.9666	—	×
NC-03 (no S2)	0.7756	—	×

Cuadro 16: Adversarial perturbation (EXP-NC-AD-01). Illustrative comparison: NC-01 and NC-02 deflect less deeply than `vsf-nucleus` but fail to recover; NC-03 deflects to a comparable depth and also fails to recover. Only the fully governed kernel completes the dip-and-recovery cycle.

Variant	Mean `stasis_cos`	σ
`vsf-nucleus` (reference)	1.0000	< 0.001
NC-01 (S5 off)	1.0000	< 0.001
NC-02 (trust const.)	1.0000	< 0.001
NC-03 (no S2)	0.874	0.003

Cuadro 17: Long-run basin stability (EXP-NC-LR-01, 110 min). Illustrative comparison: NC-01 and NC-02 occupy the same basin as `vsf-nucleus`. NC-03 without S2 coordination settles at a distinct attractor centred at 0.87.

EXP-NC-LR-01. NC-01 and NC-02 converge to the same basin as `vsf-nucleus` (stasis_cos = 1.0000, $\sigma < 0.001$). NC-03, with S2 coordination disabled, stabilises at a distinct attractor: stasis_cos $= 0.874 \pm 0.003$. The VSM blocks A–G collectively determine the attractor basin; S5 and trust govern the recovery path within that basin.

Variant	Duration	Dips	Dip floor	`stasis_cos` (final)
`vsf-nucleus` (reference)	180 min	∼20/cycle	0.942	1.0000
NC-01 (S5 off)	180 min	0	—	1.0000
NC-02 (trust const.)	180 min	0	—	1.0000
NC-03 (no S2)	180 min	88,727	0.771	1.0000

Cuadro 18: Cyclic thermal load (EXP-NC-TC-01, 180 min per variant). Illustrative comparison: the dip floor 0.942 is not reproduced by any ablated variant – it is an exclusive signature of the complete VSF architecture. NC-01 and NC-02 produce zero dips; NC-03 produces unstructured dips at a different floor.

EXP-NC-TC-01. The critical finding is the *exclusivity* of the dip floor 0.942: no ablated variant reproduces it. NC-01 and NC-02 show zero dips under identical thermal load, confirming that the dip is an algedonic signal originating from S5 governance, not a substrate thermal response. NC-03 produces 88,727 unstructured dips at floor 0.771: S2 coordination is required to organise the response into the fixed mode observed in `vsf-nucleus`.

15.4. Findings

B1 – confirmed. NC-01 (S5 off) produces zero dips under cyclic thermal load. The dip is an S5 algedonic signal, not a substrate response.

B2 – confirmed. NC-02 (trust const.) also produces zero dips. Trust alone cannot generate the algedonic response; the S5 governance structure is the primary driver.

B3 – confirmed. NC-03 (no S2) produces 88,727 unstructured dips at floor 0.771. The fixed mode at 0.942 requires S2 coordination.

B4 – confirmed. No NC variant recovers after adversarial perturbation. `vsf-nucleus` is the only configuration exhibiting the complete dip-and-recovery cycle.

$B_{\varnothing}$ – **rejected.** No NC variant reproduces dip_floor = 0.942. The value is an exclusive signature of the complete architecture.

C2b – confirmed. NC-03 (no S2) occupies a distinct attractor basin (0.874) from `vsf-nucleus` and NC-01/02 (1.000). The VSM blocks A–G collectively select the attractor basin.

Combined, these results support a **two-layer interpretation** of viability:

1. **Architecture** (VSM blocks A–G) selects the attractor basin. NC-03 without S2 settles at basin 0.87; all other variants occupy basin 1.0, regardless of governance state.
2. **Governance** (S5 + trust) determines the recovery path. All NC variants remain viable (final stasis_cos = 1.0000) but cannot return to the stasis reference after adversarial perturbation; `vsf-nucleus` alone executes the full recovery cycle.

The OU substrate has intrinsic mean-reversion that preserves viability in all variants. VSF adds *structure*: the dip floor becomes a fixed mode, the recovery becomes a governed trajectory, and both become observable and certifiable metrics rather than unlabelled displacements in an unlabelled space.

16. VSL: Semantic Compression as Inter-Layer Protocol

Question

Can a compact formal language preserve operational semantics across system boundaries with significantly fewer tokens than natural language?

16.1. Motivation

The experiments described above required communicating complex protocol specifications – thermal regimes, stability thresholds, governance axioms – between system layers and across agent boundaries. Natural language instructions are verbose, ambiguous, and sensitive to paraphrase; they do not compress cleanly.

Viable System Language (VSL) is a machine-readable notation derived from Beer's VSM formalism. A VSL instruction expresses system hierarchy, axioms, state transitions, and metrics in a grammar that can be parsed without prose disambiguation. We measured whether this compactness is real and whether it survives cross-agent transmission.

16.2. Protocol

Five protocol specifications of increasing complexity were written twice: once as natural-language prompts (`prompt_A`), once as VSL instructions (`instruccion_B`). The B versions were further split into two variants: `B_full` (standalone – recipient has no prior schema) and `B_schema` (machine-to-machine – recipient has VSF schema pre-loaded). Token counts used `tiktoken` (`cl100k_base`).

16.3. Resultados

Standalone VSL achieves 38–51 % compression. With a shared schema (machine-to-machine mode), compression rises to 72–79 % and grows with protocol complexity, consistent with the original 83 % observation on a single case.

Cuadro 19: VSL compression results (tiktoken cl100k_base). r_{full} = standalone compression ratio; r_{schema} = machine-to-machine compression ratio.

Case	T_A	$T_{B,\text{full}}$	$T_{B,\text{schema}}$	r_{full}	r_{schema}
`pvsnp`	328	161	–	0.51	–
`thermal_senescence`	638	346	–	0.46	–
`spec_boot+Lyapunov`	717	432	148	0.40	0.79
`proxmox+S4`	698	411	198	0.41	0.72
`PSI/1.0`	641	397	147	0.38	0.77

16.4. Semantic drift test

The five `B_schema` instructions were submitted to an independent language model (Gemini) with the VSF schema pre-loaded. Structural fidelity – correct identification of S-layer hierarchy, $\varphi\uparrow$ / $\downarrow$ channels, PSI thresholds, covenant verification step, and metric names – was $\approx 100\,\%$.

Semantic fidelity was $\approx 40\,\%$: the model systematically substituted its own operational context for VSF domain referents (e.g. $\varphi\uparrow$ checkpoints mapped to cloud-sync operations; `vm-nucleus` mapped to an internal knowledge module). This *context substitution* is structurally silent – the parse tree is correct but the referents are wrong.

The fix is a lightweight domain anchor:

```
domain_context: {target=vsf-nucleus, env=Rust/LXC, layer=S1}
```

added to the `B_schema` header. This adds ≈ 8 tokens and eliminates context substitution in subsequent tests.

Cross-model replication. As an independent replication, the VSL v1.0 specification was provided to Qwen3.6-Plus (a distinct architecture and training distribution from Gemini and Claude). Without additional instructions, the model parsed the grammar and produced a syntactically valid `spec.vsm` document following all VSL conventions. This constitutes a third independent model confirming structural fidelity, consistent with the hypothesis that VSL grammar is learnable from the specification alone.

16.5. Hypotheses result

H1 – confirmed. Schema mode: 72–79 % $\geq$ 70 % threshold.

H2 – partial. Structure preserved; domain grounding requires the anchor block.

H3 – split. Structural drift $\approx 0\,\%$; semantic drift $\approx 60\,\%$ without anchor, $\approx 5\,\%$ with anchor. The original single metric conflated two independent properties.

16.6. Implication

VSL compression enables governance signals to cross organisational boundaries – between kernels running on different hardware, maintained by different teams, implemented in different languages – at a token cost that scales with protocol complexity rather than prose verbosity. The structural/semantic drift distinction introduced here is a prerequisite for any formal correctness guarantee of inter-agent communication.

17. Sustrato Físico como Capa de Gobernanza

The cgroup v2 hierarchy imposes physical limits on each S1 kernel: `cpu.weight` $\propto$ trust, thermal throttling caps available compute. These constraints are not policy choices – they are hardware-

enforced boundaries.

In VSM terms: the hardware substrate acts as a physical S5. Policy can be overridden by a privileged process; the thermal ceiling cannot. The experiments reveal three governance layers operating simultaneously: (1) software policy (covenant axioms, trust allocation), (2) cgroup v2 limits (S3's allocation signal), (3) hardware throttling (the physical floor).

The same mathematical structure – a feasibility constraint that cannot be bypassed – appears at all three levels. This is the sense in which the physical substrate *is* a governance layer, not merely an implementation detail.

18. Preguntas Abiertas

The following questions were identified prior to experimentation; their current status is indicated in each item.

Q1 – Thermal response curve (resolved) The `thermal_ladder` experiment (Section 10) confirms H_{floor}: the attractor degrades by only $\Delta\cos = 0.0023$ across a 30 °C range under transient stress. The damage function is $T \times t$, not T alone (Equation 15).

Q2 – Thermal senescence (resolved for constant-H_{rest} regime) Does the attractor drift under *sustained* thermal stress over hours? Two full 240-minute runs at different temperatures answer this: EXP-TSM-01 at 45 °C (609 M ticks, stasis_cos = 1.0000, soplos = 0) and EXP-TSM-65C at 65 °C (383 M ticks, stasis_cos = 1.0000, soplos = 0). Both results are identical: no drift at any tested sustained temperature. The Lyapunov senescence term (δ) is structurally zero with the current constant-H_{rest} implementation (Section 9.4). **Open**: genuine senescence requires adaptive H_{rest} (planned as EXP-SEN-01); Q2 is closed for the present implementation.

Q3 – Basin floor under extreme perturbation (EXP-AB-01, 2026-04-27) A parameter sweep across $\delta = [1.0, 1.5, 2.0, 3.0, 5.0, 10.0]$ using an impulsive ground-state kick (10-tick impulse toward the minimum-entropy configuration, then free Boltzmann relaxation) established an empirical floor of $\cos_{\text{floor}} \approx 0.875$, saturating for $\delta \geq 3$. The viable boundary ($\theta_{\text{viable}} = 0.70$) was not reached at any δ, giving a resilience margin of $0.875 - 0.70 = 0.175$. H2 (viable breach) is **rejected** for the impulsive-kick perturbation family; H4 (finite recovery time τ_{rec}) is confirmed for $\delta \geq 3$ ($\tau_{\text{rec}} = 991$–999 t). No governance intervention was required at any δ. **Open**: whether sustained or multi-variable perturbation families can reach the boundary; proof of a formal lower limit on Ω.

Q4 – Full Lyapunov spectrum The per-run tracker reports D_{eff}, computed as active_dims + trust × 2.0 – a heuristic proxy, not the formal Kaplan-Yorke dimension (which requires the full Lyapunov spectrum via Benettin's method or equivalent). A full spectrum requires two instances diverging from nearly identical initial conditions; this measurement has not been made (planned as EXP-LY-01).

Q5 – Multi-kernel attractor interaction Four kernels converge to the same basin (Section 6). What happens at $N > 4$ under shared thermal budget? Is there a recursion threshold where `viable_all` < 1?

Q6 – Trust dynamics Trust oscillates between 0.000 and 0.002 in long runs, not the 0.8–1.0 range of production operation. The trust accumulation/decay balance in `spec_boot` may not represent the trust dynamics of a kernel under real workloads. The relationship between `spec_boot` trust and deployment trust requires clarification.

Q7 – L_3 reachability: formal lower limit on Ω EXP-AB-01 (Q3) establishes an empirical floor at $\cos \approx 0.875$ for impulsive-kick perturbations. The theoretical question is whether a formal lower bound on Ω can be proved under arbitrary perturbation – not just the impulsive family tested. A proof would require showing that the attractor's Lyapunov function $V(\mathbf{x}) = 1 - \cos\theta$ has a finite escape time lower bound for all admissible perturbations; this has not been attempted. **Open**: non-impulsive families (sustained push, coordinated multi-variable, adversarial scheduling) and formal proof of $\inf_\Omega \cos\theta > \theta_{\text{viable}}$.

19. Limitaciones

Single hardware node All experiments are conducted on n02 (Intel Core i7-7600U, 2 cores / 4 threads). Thermal dynamics, scheduling latency, and cgroup behaviour may differ on other architectures. Reproducibility on ARM, RISC-V, and multi-socket systems has not been tested.

Synthetic workload The thermal stress experiments use a synthetic floating-point loop (cosine similarity over random vectors) as the heat source. A real workload (database queries, inference, compilation) would produce different Block B dynamics and different attractor trajectories.

Constant entropy ceiling As noted in Section 9, H_{rest} is a compile-time constant. The Lyapunov senescence term is therefore structurally zero, and any claim about long-run attractor stability under genuine entropy drift is not yet testable with the current implementation.

State vector dimensionality The $\mathbb{R}^{128}$ state vector has seven semantic blocks, but many dimensions (Block G in particular) are underpopulated in current experiments. The measured geometry reflects the active dimensions; the full 128-D geometry remains unexplored.

No external environment All experiments are closed-system runs. A viable system, by Beer's definition, must respond to environmental change. The interaction between `vsf-nucleus` and an environment generating novel demands has not been measured.

Proof sketch, not full proof Proposition 1 and Theorem 2 are stated with proof sketches. Brouwer's fixed-point theorem requires compactness of Ω, which depends on the consistency and boundedness of the axiom set – assumptions that have not been formally verified for the full axiom system.

Architecture ablation scope Experiment 11 tests three specific ablation conditions: S5 disabled (NC-01), trust held constant (NC-02), and S2 coordination removed (NC-03). Many other ablation combinations – simultaneous multi-layer removal, partial degradation, adversarial governance configuration, or layer-by-layer reintroduction – have not been tested. The three-condition result is sufficient to support the two-layer viability claim, but a systematic ablation programme spanning the full combinatorial space of VSM blocks would strengthen it considerably.

Proxy metrics Both principal observables are proxies. D_{eff} (Section 5) is a heuristic (active dims + trust $\times 2$), not the formal Kaplan-Yorke dimension. `stasis_cos` (Equation 2) is standard cosine similarity, not a complete geometric description of the attractor.

Alignment: single observation, not systematic validation The Discussion hypothesis (§20) – that geometric character could serve as an internal coherence indicator – rests on a single

experimental condition (EXP-TC-01, one thermal profile, one hardware node). It is a hypothesis derived from an observation, not a validated result.

20. Discusión

20.1. El retrato empírico del atractor

Across eleven completed experiments, including three negative-control ablations, a consistent picture emerges. The attractor is:

- **Compact**: 18.2° cone, 99.4 % variance in PC1+PC2.
- **Unique**: different initial conditions converge to the same basin (mean pairwise cosine 0.9854).
- **Thermodynamically stable**: viable at 68 °C, min cosine 0.9365, recovery during hold phase.
- **Intrinsically viable**: viability does not require S3 coordination.
- **Deeply welled**: perturbation $\delta = 1.0$ pushes cosine to 0.79 but does not breach viability.
- **Dimensionally stable**: $D_{\text{eff}} = 3.00$ from 10^3 to 6×10^7 cycles.
- **Elastically resilient**: under cyclic thermal load (EXP-TC-01), perturbations land at a *fixed dip floor* (stasis_cos = 0.942, invariant across 20 cycles) and recover in $\tau_{\text{rec}} < 30\,\text{s}$ – one dip per thermal cycle, zero accumulation across 60 minutes.
- **Architecturally stratified**: negative controls confirm a two-layer structure – VSM blocks A–G select the attractor basin; S5 governance determines the recovery path. The dip floor 0.942 is an exclusive signature of the complete architecture.

Taken together, these properties describe a system whose viability is encoded in the geometry of its state space rather than in any particular state, and whose resilience is a structural consequence of its architecture rather than an externally enforced condition. This is what it means, empirically, for a system to be viable in Beer's sense.

20.2. Identidad geométrica bajo perturbación

The recovery time τ_{rec} (Equation 17) introduces a qualitatively new observable: not how far the system deflects, but how it deflects. The constant dip floor observed in EXP-TC-01 – 0.942 in all 20 cycles without variation – is evidence of a *fixed perturbation mode*: a preferred direction of deflection in the 128-dimensional state space under this load profile.

This is structurally analogous to what Lindblad (1976) calls a *jump operator*: the dominant dissipative channel through which a quantum open system loses coherence to its environment. The fixed dip direction in $\mathbb{R}^{128}$ could be interpreted as the classical analogue of a leading jump operator, with the thermal phase transition acting as the environment. Whether this analogy extends to a quantitative Lindblad formalism is an open question; the exploration is left for future work.

These results are **compatible with** the hypothesis that a geometric signature – the shape of a system's response to perturbation – could serve as an internal coherence indicator, complementary to the external approaches (output inspection, log audit, declared-intention review) that prevail in current practice. A system with a stable perturbation mode – one that deflects predictably and returns reliably – exhibits what might be called *geometric character*: behavioural regularity that is not enforced from outside but emerges from structure.

The negative controls (Experiment 11, §15) provide the required verification. NC-01 (S5 off) and NC-02 (trust constant) produce zero dips under identical thermal load: the fixed mode does not emerge from substrate dynamics alone. NC-03 (no S2) produces 88,727 unstructured dips at a different floor (0.771), without the fixed mode. Only the complete architecture reproduces the dip floor at 0.942, invariant across cycles. The constant dip floor is therefore an architectural signature, not a substrate coincidence – which upgrades the geometric character hypothesis from *compatible with the data* to *empirically supported by ablation.*

20.3. Arquitectura como gobernanza, no producción

A natural objection arises: if ablated variants NC-01 (no S5) and NC-02 (constant trust) also remain viable after adversarial perturbation, does that render the VSF architecture redundant? Experiment 11 (§15) answers this empirically.

Practical validation (Phase 1.5): The scheduler framework `thermal_scheduler_enhanced.py` implements a real-world instance of this principle: when thermal sensors are unavailable (virtualized environments lack hwmon drivers), the system enters SAFE_MODE – a graceful fallback that pauses new work and allows in-flight operations to complete. This is an instance of $A_{\text{SafeFallback}}$: missing telemetry triggers conservative operation, not failure. The architecture remains viable even when critical observables (thermal data) become unavailable, confirming the layered governance model predicted here.

The objection conflates *achieving* viability with *operating* viability. The StateVector $\mathbf{x} \in \mathbb{R}^{128}$ has intrinsic mean-reversion dynamics – an Ornstein-Uhlenbeck-like process that may produce bounded trajectories independently of S5 policy. VSF does not alter this underlying geometry; it governs *which operations execute*, *when*, and *under what policy constraints.* The analogy is precise: combustion dynamics are substrate-intrinsic to any engine; the driver does not change thermodynamics but operates the system toward a declared goal. A null result in the adversarial control experiment would demonstrate that the StateVector has intrinsic resilience – and VSF is the framework that makes that resilience *observable, certifiable, and intentional.*

Specifically, VSF adds three layers that are absent in any uncontrolled baseline:

- **Observability**: stasis_cos, D_{eff}, and τ_{rec} are VSF-constructed metrics; a substrate process has no analogue.
- **Intentionality**: the governed flag, trust feedback, and S5 lifecycle ceiling encode a policy that the system operates *toward* viability, not merely within it.
- **Certificability**: the system can formally declare its viability state – a prerequisite for autonomous governance and inter-system trust.

A parallel consideration applies to the fixed dip floor. Even if the perturbation mode direction in $\mathbb{R}^{128}$ is substrate-determined – a property of the hardware's thermal response rather than the governance layer – the *characterisation* of that dip is VSF-constructed. The stasis reference vector $\bar{\mathbf{s}}$ is itself a VSF artifact: the running mean of governed observations under S5 policy. The dip floor value 0.942 is meaningful only relative to $\bar{\mathbf{s}}$; without that reference, the perturbation is an unlabelled displacement in an unlabelled space. The distinction is precise: the substrate *produces* the phenomenon; VSF makes it *intelligible.* Experiment 11 resolves this empirically: NC-01 and NC-02 produce zero dips under identical thermal load, and NC-03 produces unstructured dips at a different floor (0.771). The floor 0.942 is governance-dependent, not substrate-invariant.

20.4. Theory preceding experiment

The predictions of Sections 3–4 were made before running the experiments. No post-hoc adjustment was made. The confirmation of $\overline{dL} \leq 0$, the narrow attractor cone, and `viable_full =`

`true` throughout thermal stress provides evidence that the theoretical framework captures real dynamics.

20.5. Worker pool as S3→S1 prototype

The persistent worker pool pattern of `thermal_stress_v3` (Barrier + AtomicU64 quota) is structurally isomorphic to the S3→S1 coordination protocol in a multi-kernel deployment. Each worker corresponds to one S1 kernel; the quota to the resource allocation signal $\phi^{\downarrow}$. The `cgroup_multikernel` experiment directly tests this: four S1 kernels, S3 adjusting `cpu.weight` per thermal pressure, `viable_all = 100%`.

21. Conclusión

This paper documents an ongoing experimental programme investigating the geometric properties of a viable system kernel governed by cybernetic axioms.

The core theoretical result – that axiom-governed dynamics produce a compact attractor in $\mathbb{R}^{128}$, certified by a Lyapunov function – is supported by eleven hardware experiments, including three negative-control ablations. The attractor is compact, unique, thermodynamically stable, and intrinsically viable. The negative controls confirm a two-layer structure: VSM blocks A–G select the attractor basin; S5 governance determines the recovery path. The fixed dip floor (0.942) is an exclusive signature of the complete architecture – an algedonic signal of S5 in action, not a substrate artefact. The stability intervention threshold (θ_{PSI}) has been revised from a failure indicator to a precautionary support level, following the empirical finding that non-viability appears to be structurally unreachable under normal perturbation.

What remains open is at least as interesting as what has been confirmed. The thermal response curve has been characterized (EXP-TL-01): degradation depends on $T \times t$, not temperature alone. Genuine senescence has not been demonstrated and may require an adaptive entropy ceiling. The full Lyapunov spectrum has not been measured. The basin floor under impulsive entropic perturbation is now empirically bounded at $\cos_{\text{floor}} \approx 0.875$ across six δ values (EXP-AB-01), a resilience margin of 0.175 above the viable threshold; sustained and multi-variable perturbation families remain to be tested.

This is a system that can maintain its identity while adapting to environmental change – Beer's definition of a viable system. Whether the geometry of that identity is as rich and robust as the current experiments suggest, or whether the open questions will reveal qualitative failures, is the work ahead.

22. Future Work and Broader Programme

This work documents a single viable system kernel in isolation. The implications extend across multiple research directions, pursued in parallel but separate from the core paper.

Measurement and control

The full Lyapunov spectrum (work item §18 Q4) requires eigenvalue computation via Benettin's method on the Jacobian of f_0. This will replace the D_{eff} heuristic with the formal Kaplan-Yorke dimension. Similarly, richer geometric metrics (Hausdorff distance, Wasserstein metric) may provide finer characterization than cosine similarity.

Environmental interaction

Beer's original definition of viability requires the system to adapt to environmental change. The closed-loop experiments reported here are the foundation; coupling `vsf-nucleus` to a stochastic environment model (e.g., arrival processes, resource contention) will test whether adaptation is observable and whether the geometric signature remains stable under truly novel demands.

Architectural extension: vOS

The immediate engineering extension is to scale `vsf-nucleus` into a minimal operating system (vOS) running on bare metal, with Ollama/Qwen as the S4 cognitive layer. The roadmap includes support for multi-kernel coordination (Proxmox virtualisation), persistent state storage (PostgreSQL), and asset graphs (Neo4j) to support the full VSF stack in a production setting. This is anticipated for Q4 2026 pending Phase 1.5 integration results.

Governance and ethics (A0 covenant)

The ethical axioms A0/A0.1 are currently embedded at compile time. A separate programme explores the mathematics of axiomatic governance in multi-agent settings: how to verify that axioms remain unviolable under adversarial perturbation, how to extend them to coalitions, and how they relate to formal theories of alignment. This work involves collaboration with theoretical computer scientists and ethicists and is documented in a separate forthcoming manuscript.

Connection to open system thermodynamics

The fixed perturbation mode (§20.3, the dip floor at 0.942) bears structural resemblance to the jump operators of Lindblad's theory of open quantum systems. Whether this analogy extends to a quantitative formalism – e.g., interpreting the thermal phase transition as an environmental channel – is an open question. Preliminary exploration suggests the connection is more than analogy; formal development is deferred to a dedicated paper on classical-quantum analogues in cybernetic systems.

Code availability. The kernel implementation (`vsf-nucleus`) and all experimental tools are available upon request from the corresponding author (`luis@tallerpineda.cl`).

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A. Implementation Notes

A.1. Kernel architecture

`vsf-nucleus` is implemented in Rust (stable, 2024 edition). Relevant modules:

- `src/state_vector.rs` – $\boldsymbol{x} \in \mathbb{R}^{128}$, blocks A–G; `viable()`, `lyapunov()`, `cosine_similarity()`.
- `src/vmem.rs` – vector memory V_{mem}, `insert()`, `query_nearest()`, `merge_similar()`.
- `src/attractor.rs` – `LyapunovTracker`, `StasisVec`, convergence detection.
- `src/s4.rs` – S4 cognitive client (Ollama/Qwen), covenant embedding, $\phi^{\downarrow}$ channel.
- `src/main.rs` – experiment dispatch, telemetry.

A.2. Covenant embedding

The governance covenant is embedded in the binary at compile time via `include_str!()` and verified by SHA-256 before every S4 call. This makes the ethical axioms A0/A0.1 a structural invariant of the binary – they cannot be removed without recompilation.

A.3. Persistent worker synchronisation

The shutdown-safe Barrier pattern used in `thermal_stress_v3`:

```
loop {
    start_barrier.wait();
    let stop = stop_flag.load(Relaxed);
    if !stop {
        let n = quota.load(Relaxed);
        let mut acc = 0.0f64;
        for _ in 0..n { acc += sv.cosine_similarity(&st); }
        black_box(acc);
    }
    done_barrier.wait();    // unconditional -- prevents coordinator
       deadlock
    if stop { break; }
}
```

A.4. Experiment specifications

All experiments are formally specified in `docs/experiments/<name>/spec.vsm` and results are recorded in `results.vsm` in the same directory. The master results index is at `docs/experiments/results_index.vsm.md`.

A.5. Reproducibility and logical sequencing

Each experiment is specified using a formal gate-based notation: logical conditions (IF condition THEN action) rather than temporal estimates (X will take Y minutes). All event timestamps are recorded in ISO 8601 UTC format (`started_at`, `completed_at`, `checked_at`). This enables:

- Audit trails (who did what when)

- Causal ordering (event A caused B iff A.completed_at < B.started_at)
- Reproducibility checks (actual durations vs baselines)

No durations are estimated; observed durations are compared against historical runs to detect anomalies or improvements. This methodology avoids false confidence from invented timelines and provides a clear trigger path for future experiments: `IF all gates pass THEN proceed to next phase`.